

Utility comparison and calibration targets

September 23, 2026 | Prepared for the 10:00 EDT review

Summary

The overnight experiment produced 444 scored evaluations and 2,571 completed stationary solves. All four starting smokes passed. Twenty of the 40 search workers finished normally; twenty stopped at a numerical gate after preserving their earlier results. The results are now collected and reviewed. This was a substantial finite search, not a converged calibration.

My working recommendation remains one persistent earnings process plus the age profile, with heterogeneous initial wealth. Keep the housing floor as the comparison baseline for now, but do not adopt either utility specification from this run. The floor achieves a better overall fit under the frozen objective; child-dependent shares improve ownership in the main comparison but create larger misses elsewhere. Neither delivers a satisfactory housing fit.

The target review largely reproduces the existing empirical numbers. The priority is a narrow measurement decision: align birth timing and the first-birth housing comparison in the model and data. Do not replace a hard target simply to improve fit. The alternative moments below remain proposals, and the overnight jobs used the original targets and inherited wealth throughout.

Selected points: all 13 target moments

B uses one persistent earnings process; D adds iid risk. All cells retain heterogeneous entry wealth. Each value is from the lowest-loss verified smoke or production case. Repeats are excluded from selection. These finite-search results are not a convergence certificate.

Target moment	Target	B_floor	B_shares	D_floor	D_shares
Initial fertility normalization	2.1	2.1005	2.1	2.1001	2.1002
Childless women, ages 40-44	0.19828	0.16909	0.1496	0.1628	0.12597
Exactly one child among mothers, ages 40-44	0.21366	0.24629	0.25282	0.25062	0.28478
Period mean first-birth age	25.976	26.658	25.727	26.409	26.246
First births at age 30+	0.24928	0.26562	0.21398	0.24933	0.24595
Wealth / annual gross labor earnings	6.1459	5.2412	6.8694	5.4899	5.1972
Annual bequests / aggregate wealth	0.0088	0.006829	0.0069287	0.0061056	0.0063029
Old wealth/income p90 / median, ages 76-84	3.5159	4.0994	3.1184	4.3821	4.3755
Mean occupied rooms, capped at 9	5.5611	6.4767	7.0882	6.5219	6.1667
Ownership, heads 30-55	0.64833	0.39493	0.51325	0.44551	0.36679
First-birth room response, -1 to +3	0.72025	0.98725	1.0437	1.1473	0.37748
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.28295	0.49163	0.29649	0.54813
Recent-parent ownership gap	0.1629	0.12501	0.055075	0.13583	-0.017945

Common smoke anchors: all 13 target moments

These are the cell-specific smoke_01 anchor cases copied by the collector. Missing cells remain unavailable. A smoke anchor is a checked starting point, not a search result.

Target moment	Target	B_floor	B_shares	D_floor	D_shares
Initial fertility normalization	2.1	2.1	2.1002	2.1	2.1002
Childless women, ages 40-44	0.19828	0.18531	0.10477	0.17346	0.10581
Exactly one child among mothers, ages 40-44	0.21366	0.24344	0.2988	0.2504	0.29771
Period mean first-birth age	25.976	27.899	25.909	27.248	25.866
First births at age 30+	0.24928	0.33875	0.22527	0.29857	0.22325
Wealth / annual gross labor earnings	6.1459	6.2005	6.1835	5.9305	6.0096
Annual bequests / aggregate wealth	0.0088	0.006216	0.0064419	0.0063782	0.0065333
Old wealth/income p90 / median, ages 76-84	3.5159	3.7337	3.5745	3.8304	3.6906
Mean occupied rooms, capped at 9	5.5611	6.2355	6.1416	6.2295	6.1327
Ownership, heads 30-55	0.64833	0.45893	0.38644	0.45265	0.38994
First-birth room response, -1 to +3	0.72025	1.3183	0.57467	1.3398	0.57441
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.3114	0.55697	0.30945	0.57246
Recent-parent ownership gap	0.1629	0.1138	-0.03347	0.12579	-0.026429

Collection and verification status

Collector inventory: 736 planned cases; 464 attempted; 4 selected cells. This status describes collection, not economic adequacy.

Cell	Verified scores	Rejected	Failed	Running/started unresolved	Incomplete	Unrun	Collection rejected
B_floor	193	0	9	0	0	126	0
B_shares	192	0	10	0	0	126	0
D_floor	36	0	0	0	0	4	0
D_shares	23	0	1	0	0	16	0
Total	444	0	20	0	0	272	0

Selected-point repeat receipts

Runner-reported completions, collector-scored repeats, and raw finalizer receipts are separate evidence. A raw finalizer statement is not independently revalidated here. File-based running or started states are unresolved without separate scheduler or process confirmation.

Cell	Runner-reported completed	Collector-verified repeats	Finalizer receipt status
B_floor	0	0	not present
B_shares	0	0	not present
D_floor	2	2	not present
D_shares	0	0	not present

Scientific verification status

Independent saved-checkpoint comparison verifies the ORIGINAL selected D_floor point against BOTH repeats: price, finite V/g array bytes, all finite native moments, early/fiscal/budget measurements, psi, normalization, full 13 targets, 17 parameters and loss agree exactly. Seven NaN locations agree as well: they are in unused permanent-income-type diagnostics, absent because this experiment has no fixed types. Frozen solver/calibration source confirms these explicit missing defaults. The raw finalizer nevertheless FAILED during strict JSON serialization of those NaNs, and that receipt is preserved. This is a separate reviewed comparison, not a repaired or successful raw finalizer. B_floor, B_shares and D_shares have no two-repeat verification.

Shared smoke parameters verified from plans

The eight listed structural coordinates are identical in all four smoke plans. Utility specifications still differ in their remaining structural dimensions: floor cells search h_P, while child-dependent-share cells search two share tilts. Fertility utility scale psi is normalized separately and is not included among these fixed coordinates.

Shared coordinate	Smoke anchor value
H0	7.5796
beta_annual	0.98614
chi	1.0094
first_birth_fixed_cost	0.31519
kappa_fert	0.29331
kappa_fert_continuation	0.45387
theta0	0.040247
theta1	0.092348

Recommendations

Keep persistent income plus the age profile and heterogeneous entry wealth as the working direction. The simple-income floor case is the current comparison baseline, not an adopted paper calibration. Supplementary persistent+iid results do not establish that extra income risk is needed; they use a different numerical approximation and much less search.

Decide the within-period birth clock and implement the corresponding model/data housing observer together before the next long refit. Preserve audited wealth and housing targets unless their economic definition is explicitly changed. National ownership is higher than the current target, so changing geography cannot explain away low model ownership.

Before adopting a utility specification, complete the six missing selected-point repetitions, selected-point income/wealth-resolution and continuation-value checks, and diagnose the ownership-versus-rooms tradeoff using the retained plots. These tasks were not silently rerun. Lower weighted loss is insufficient: the share model has one extra search coordinate and neither search is proven optimal.

Consider the common-income-scale entry law only as a separately reviewed candidate. Its units and current renter budgets are checked, but the joint distribution, tail discretization and dynamic feasibility remain untested. The linked-minor-child room gap and wealth-level dispersion are also candidates, not overnight target replacements.

Limitations

The objective mixes sampling uncertainty, across-year variation and an external bequest tolerance. Its numerical value is a diagnostic distance, not a chi-squared test or evidence of efficient SMM weighting. Twelve scored moments versus nine or ten searched coordinates is a counting check, not proof of identification.

Only the persistent+iid floor selection has both exact repetitions. The other three selections have no completed repeat receipts because their production arrays contained failed workers. All four selected checkpoints pass the original scoring gates; this does not establish global numerical reliability or grid convergence.

All selected entry distributions preserve the intended wealth marginal without censoring. Very small all-age masses still have deeply negative saved values; small population mass does not bound their effect on continuation choices. Four-year information timing, stationary risk at age 18 and household-grid robustness remain explicit limitations.

The standard diagnostic images are retained unchanged. Some source legends overlap plotted lines; the 21-state wealth figures clip the last legend entries within the source PNG. Full-domain wealth-policy axes also compress the occupied range. These source readability limits are disclosed; no ad hoc graph replaces the agreed set.

Experiment and actual execution

40 production workers: 18 per utility with 15-state persistent earnings, plus 2 per utility with 7-by-3 persistent+iid earnings. Each arm has its own two-case smoke, search, two selected repetitions and final comparison. No zero-entry fallback is allowed.

Named reference: the inherited-wealth B/D design in the September 22 earnings/entry battery. The overnight floor arm retains that utility; the shares arm changes its housing preference as stated below. All arms retain heterogeneous entry wealth, the age profile, current-income purchase eligibility, transfers, four-year timing, equivalence scale and target/weight system. Wealth remains 160 nodes up to 3000; persistent income uses the previously tested 15-node refinement rather than 7 nodes, while persistent+iid stays at 21. The search recalibrates 9 floor or 10 share coordinates, with fertility utility normalized separately.

The utility difference is experimental. The floor imposes an additional housing requirement only when children live at home. The alternative removes that requirement and lets the housing share of expenditure rise with current dependents. Both retain the existing equivalence scale. No manuscript specification has been adopted.

Search and verification have terminated, with 464 attempted objectives out of 736 planned: 444 scored, 20 failed and 272 unrun. Stationary solves: 2,590 started, 2,571 completed, 19 unfinished. The 19 equilibrium failures stopped incomplete initial solves; the budget-rejected case completed its numerical solves but was not scored. No retry, gate relaxation or extra computation was launched after collection.

Nineteen worker failures were the unchanged initial housing-equilibrium strict gate. One worker failed the exact-zero budget-excess-mass test: mass $1.47\text{e-}24$, maximum occupied spending excess $1.42\text{e-}14$. The mass test remained binding even though the latter is below the $1\text{e-}9$ tolerance. These failures are recorded as failures; they are not evidence that the economic specification is impossible.

Only the supplementary floor chain completed both repeats. A NaN in unused fixed-type diagnostic fields prevented its finalizer receipt from serializing. Independent comparisons of the selected point with both saved repetitions pass as described above. Other expected repeats were not completed; one scheduler terminal row is absent, so its detailed cancellation status is not asserted. All experiment jobs are terminal in the fresh scheduler check and no experiment job remains queued.

Scored counts include smoke evaluations and, for D_floor only, its two repeats. Selection includes smoke and production only. B denotes persistent income and D persistent+iid income throughout the tables. Equal worker allocations did not yield identical search histories or equally optimized estimates.

Earnings and utility	Workers completed / failed	Scored objectives / selected loss
Persistent + floor	9 / 9	193 / 464.8522
Persistent + child shares	8 / 10	192 / 851.2713
Persistent + iid + floor	2 / 0	36 / 416.4814
Persistent + iid + shares	1 / 1	23 / 1524.0382

Economic interpretation and parameter checks

In the main persistent-income comparison, the floor predicts ownership 39.49% versus the 64.83% target, while shares predict 51.33%. Shares therefore improve this margin, but mean rooms rise from 6.48 to 7.09 against 5.56, and the recent-parent ownership gap falls from 12.50 to 5.51 percentage points against 16.29. The floor is closer on childlessness and the first-birth room response. These are different tradeoffs, not a clean utility winner.

Adding iid risk gives a lower observed floor loss, with ownership 44.55% and first-birth age 26.41. It does not remove the housing misses. The supplementary shares point gives ownership 36.68% and a negative recent-parent ownership gap. Only two workers were assigned to each supplementary arm; search coverage and grid differences prevent treating the four selected values as pure causal effects of income risk or utility.

The standard plots show low ownership through much of the lifecycle, while most owner housing demand is concentrated in the largest available owner product. Market residuals at the selected points are small. This makes the joint tenure-and-housing-size choice a concrete diagnostic priority: a clearing housing market does not guarantee plausible ownership or unit sizes. The plots suggest where to look; they do not identify the causal source of the miss.

The annual discount factors are 0.97625, 0.99000, 0.98241 and 0.97871 in B_floor, B_shares, D_floor and D_shares, respectively. Their four-year factors are 0.90833, 0.96060, 0.93146 and 0.91753. The shares/persistent point is at the actual annual upper bound 0.99; the floor points are interior. The actual floor estimates are 2.0442 and 2.1493 housing-service units, below the 2.3 bound; the floor is triggered by positive dependent children and is not a per-child increment.

The housing-share tilts in B_shares are a first-dependent jump 0.05135 and slope 0.03288 per dependent. Relative to childless housing exponent 0.267, the one-dependent exponent becomes about 0.3512. In D_shares the corresponding exponent is about 0.2872. These are utility-expenditure weights, not direct room increments or empirically portable elasticities.

H0 is a housing-supply intercept, chi scales owner services, theta0 scales bequest utility, and theta1 is a wealth shift in that utility. Fertility kappas are taste-shock scales and the first-birth cost is in normalized utility units. Their levels cannot be certified against unrelated literature coefficients. Several are flagged within 1% of very wide raw bounds; that mechanical flag is not proof the optimum binds. The full 17/19-row tables preserve all external, derived and searched values.

The income architecture has standard precedents, but four-year household earnings coefficients cannot be imported unchanged from annual wages. The maintained direct-data persistent process has rho4 0.73459 and innovation SD 0.48383; persistent+iid has rho4 0.77614, persistent innovation SD 0.43744 and iid SD 0.16499. The earlier reviewed income packet documents uncertainty and weak iid precision; no new external income coefficient was calibrated to improve these fits.

All four selected wealth marginals match their intended inherited entry marginal to $L1 \leq 2.64e-16$, and saved entry censoring is zero. All selected V and distribution arrays are finite. Population shares with saved V $\leq -1e6$ are $2.59e-11$, $9.54e-12$, $2.26e-11$ and $3.57e-11$. These checks establish preservation of the intended initial condition, not that the empirical proxy or finite continuation penalties are harmless.

All 13 current target values and working scales

There are 12 scored empirical rows plus one separate fertility normalization. Values below remain those used by the frozen experiment. A working scale is the denominator used to turn a moment gap into its loss contribution; the final column states why these scales should not all be called comparable standard errors.

The accompanying target_contract.csv gives all target values, working weights and scales, with estimator, sample, fixed effects, uncertainty and model-measurement provenance. It excludes historical model values. This table contains no new model fit.

Moment	Frozen target	Working scale	Scale origin
Mean children ever born, age 46+	2.1	-	Separate model normalization
Childless, women 40-44	0.1982788	0.00530504	Pooled CPS approximate GVF
One child among mothers 40-44	0.2136553	0.00609113	Pooled CPS approximate GVF
Mean age at first birth	25.97626	0.0845674	Across-year variation
Share of first births 30+	0.249278	0.00849226	Across-year variation
Aggregate wealth/annual gross earnings	6.145861	0.362855	Saved sampling estimate
Annual bequests/wealth	0.0088	0.00044	Externally assigned 5% tolerance
Old wealth/income ratio, p90/p50	3.515935	0.306911	Person bootstrap
Mean rooms	5.561097	0.0883812	42-metro bootstrap
Homeownership, ages 30-55	0.648334	0.0206753	42-metro bootstrap
Room response around first birth	0.7202463	0.0852601	Person-cluster regression covariance
Rooms, 3+ versus 1-2 resident children	0.3470669	0.0597052	42-metro bootstrap
Ownership gap, recent parents	0.1628955	0.00607952	42-metro bootstrap

Decisions needed before adopting a calibration

These are proposed next steps, not changes made to the submitted jobs. Correct existing measurement mismatches explicitly; do not discard a hard-to-fit moment without replacing its identifying information.

Moment	Proposed disposition
Mean children ever born, age 46+	Keep frozen; settle population and 3+ representation before adoption.
Childless, women 40-44	Retain empirical share; align age interpolation and household interpretation.
One child among mothers 40-44	Retain empirical share; same age and top-bin contract.
Mean age at first birth	Retain matched 4-year-bin diagnostic; explicitly choose within-period birth timing.
Share of first births 30+	Retain matched age threshold; inspect late birth flows separately.
Aggregate wealth/annual gross earnings	Retain audited stock/flow units; no denominator substitution.
Annual bequests/wealth	Literature value verified; recipient/estate coverage needs matching.
Old wealth/income ratio, p90/p50	Author choice: retain ratio dispersion, or jointly replace both sides with wealth-level dispersion (4.91943).
Mean rooms	Room cap implemented; population/structure coverage must match paper claim.
Homeownership, ages 30-55	Age mask implemented; choose target geography explicitly.
Room response around first birth	Rebuild chosen common-cohort contrast and matching model observer together.
Rooms, 3+ versus 1-2 resident children	Prefer linked minor-child grouping candidate 0.33622 after model-dependent mapping review.
Ownership gap, recent parents	Active model compares current new-parent flows with empty-dependent homes, including former parents; survey timing remains approximate.

First-birth room response: what the replay establishes

One 330-second replay reproduced the original estimate 0.7202463, clustered SE 0.0852601, 49,457 observations and 4,112 people. The full 900-by-900 cohort-interaction covariance was recovered. This reproduces the implementation; it does not settle the empirical comparison or its model counterpart.

The following contrasts use common cohorts at both endpoints. The uncertainty conditions on the selected cohort weights. It excludes uncertainty from estimating those weights. The omitted event-2 coefficient is set to zero only for supported reference observations. Different windows and cohort weights are different estimands, not interchangeable robustness numbers.

The current model comparison advances treated and control branches by one four-year step, permitting subsequent births. It is not literally the same regression and cohort weighting. A four-year gap between data endpoints alone is insufficient to establish an exact counterpart.

Codebooks do not support automatically treating the 100 zero-room records as missing: zero denotes a shared-room response. The only surviving raw 98 is in 1991, when 1-98 are allowed counts. This resolves a coding interpretation, not the economic plausibility of 98 rooms. Aggregate census exports completed; a subsequent metadata export failed and is preserved.

Additional saved-covariance calculations give an eight-year -4/+4 contrast of 1.24358 rooms (SE 0.12145; 25 common cohorts, fixed preperiod weights). On the 23 cohorts also observed at -2, the same-weight -4/+4 change is 1.25709, of which 0.39841 (SE 0.10105) occurs between -4 and -2. The remaining -2/+4 contrast is 0.85868 (SE 0.10092). The longer change therefore includes substantial prebirth movement; adopting it solely to bring the target closer to the model would be unjustified.

Opus initially proposed an exact -4/+4 match, then withdrew that claim after the lead challenged its timing assumptions. An explicit model-panel observation rule and matched auxiliary regression are the recommended direction. Uniform birth dates or endpoint/midpoint conventions are sensitivity assumptions, not established timing facts or mathematical bounds. Marginal event-time support does not reproduce full individual interview histories, and education controls cannot simply be dropped only on the model side.

Contrast	Cohort weights	Estimate	Conditional SE
-1 to +3	Common 27, preperiod	.81139749	.07660629
-1 to +3	Common 27, postperiod	.79775926	.07545040
-2 to +2	Common 27, preperiod	.73459943	.07562861
-2 to +2	Common 27, postperiod	.70389142	.07542286
-4 to +4	Common 25, preperiod	1.24358063	0.12144944
-4 to +4	Common 25, postperiod	1.18619706	0.12203751
-4 to +4	Shared 23, preperiod	1.25709236	0.12656790
-4 to -2	Same shared 23 / weights	0.39841451	0.10104629
-2 to +4	Same shared 23 / weights	0.85867786	0.10092342

Two concrete measurement alternatives

Linked minor children: the ACS family-room gap falls from 0.347067 to 0.336220 rooms when groups use linked children under 18. The paired difference is -0.010847, SE 0.008077 under the same 1,000 metro draws. It is a better candidate for current dependents, but the model departure process still needs a stated age interpretation.

Old-age dispersion: on the exact same PSID sample, p90/p50 of household wealth-to-family-income ratios is 3.515935 (SE 0.306911), whereas p90/p50 of wealth levels is 4.919429 (SE 0.579804). These measure different things. Constant model pensions make the two model ratios coincide; that does not justify using either data number interchangeably.

The national ownership counterpart is 67.626%, above the current 42-metro 64.833% target. A population change therefore does not mechanically fix historical low model ownership. Choose geography from the paper claim, not from the fit.

Proposed entry wealth: units checked, equilibrium not run

Use each entrant household's nonhousing net worth divided by the same-wave mean gross earnings of working reference-person households. Unlike division by own earnings, this preserves zero-earnings entrants without imposing an arbitrary denominator floor. The established 18-24 childless-renter sample remains 1,835 family-years/1,346 people, including 67 zero-earnings observations.

The proposed 15 wealth means are formed within 3 earnings terciles and 5 normalized-wealth bins. Actual cell probabilities are retained. Their weighted mean is 0.185775923 model units because mean annual gross working earnings are 1 in the actual frozen parameter object. The native annual return is 2%; working survival is 1, with retirement survival below 1.

An initial postprocessor mistakenly read source defaults rather than the serialized seed. Its 1.112608 conversion and 0.206696 wealth mean are rejected and preserved. The corrected readout uses actual parameters after both frozen utility bindings. No submitted job or inherited entry law was changed.

Inherited unsecured debt can roll over: at age 18 the renter saving floor equals $\min(b, 0)$. Thus current feasible resources depend on $R*b + Y - \min(b, 0)$, not $R*b + Y$ alone. All 85 positive node/income pairs and 119 interpolated grid/income points have positive current slack. No Bellman or equilibrium was solved for this candidate.

The 3-by-5 means retain only 20.88% of empirical wealth variance; joint dependence is approximated. The induced weighted rank correlation is 0.24089, versus 0.28250 in the empirical rows. Product coupling places 0.2774% of mass on nine combinations absent from a finer empirical rank-overlap construction. These are disclosed approximation choices, not pass/fail tests.

Checked object	Value
Actual mean annual gross working earnings	1
Proposed mean initial wealth	0.18577592
Minimum current renter slack, continuous nodes	0.082387965
Minimum current renter slack, grid support	0.071924528
Full household/continuation/equilibrium validity	Not tested for this candidate

Literature checks and what they do not settle

Kaplan, Mitman and Violante (2020, JPE) discipline initial assets and asset-income correlation empirically, along with aggregate/lifecycle wealth and ownership. This supports measuring entry conditions rather than setting them by convenience; it does not supply a four-year fertility-model entry law for us.

De Nardi and Yang (2014, EER) use the 0.0088 bequest/wealth ratio. The original Gale-Scholz working-paper construction measures transfers to children with spouse-status adjustments and excludes pension/trust wealth. Our model total positive estates need an explicit coverage bridge. Published use alone is not validation of our observer or synthetic 5% tolerance.

Sommer (2016, JME), Table 2, targets cumulative births at ages 45 (1.90) and 25 (0.80), plus children at home at age 35 (1.43). The age-25 stock disciplines timing. The earlier 2014 working paper instead targeted first-birth age 25.5. This illustrates cohort-specific measurement; neither version requires every model to normalize fertility to 2.1.

Current model normalization pools ages 46+. CPS 40-44 in 2004/2006 gives capped mean 1.856608 and a 3+ representative 3.482793, versus inherited 2024 representative 3.602359. Decide population, top-bin representation and timing together. This mismatch alone does not prove the current closure impossible.

Appendix: B_floor full selected target fit

Collector-selected stage: production; case: worker09_proposal16; loss: 464.852226. Collection verification receipt checkpoint SHA-256: 83a28e46b36e2fbc30338d366611f3ec209f0c5a68309ee4ee9fa8523b66adee.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Initial fertility normalization	2.1	2.1005	0.00045359	-	-
Childless women, ages 40-44	0.19828	0.16909	-0.02919	35,532.30	30.275
Exactly one child among mothers, ages 40-44	0.21366	0.24629	0.032632	26,952.82	28.701
Period mean first-birth age	25.976	26.658	0.68159	139.83	64.959
First births at age 30+	0.24928	0.26562	0.016337	13,866.07	3.7009
Wealth / annual gross labor earnings	6.1459	5.2412	-0.90463	7.5951	6.2155
Annual bequests / aggregate wealth	0.0088	0.006829	-0.001971	5,165,289.26	20.067
Old wealth/income p90 / median, ages 76-84	3.5159	4.0994	0.58344	10.616	3.6139
Mean occupied rooms, capped at 9	5.5611	6.4767	0.91556	128.02	107.31
Ownership, heads 30-55	0.64833	0.39493	-0.2534	2,339.36	150.21
First-birth room response, -1 to +3	0.72025	0.98725	0.267	137.57	9.8072
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.28295	-0.064115	280.53	1.1532
Recent-parent ownership gap	0.1629	0.12501	-0.037884	27,055.82	38.831

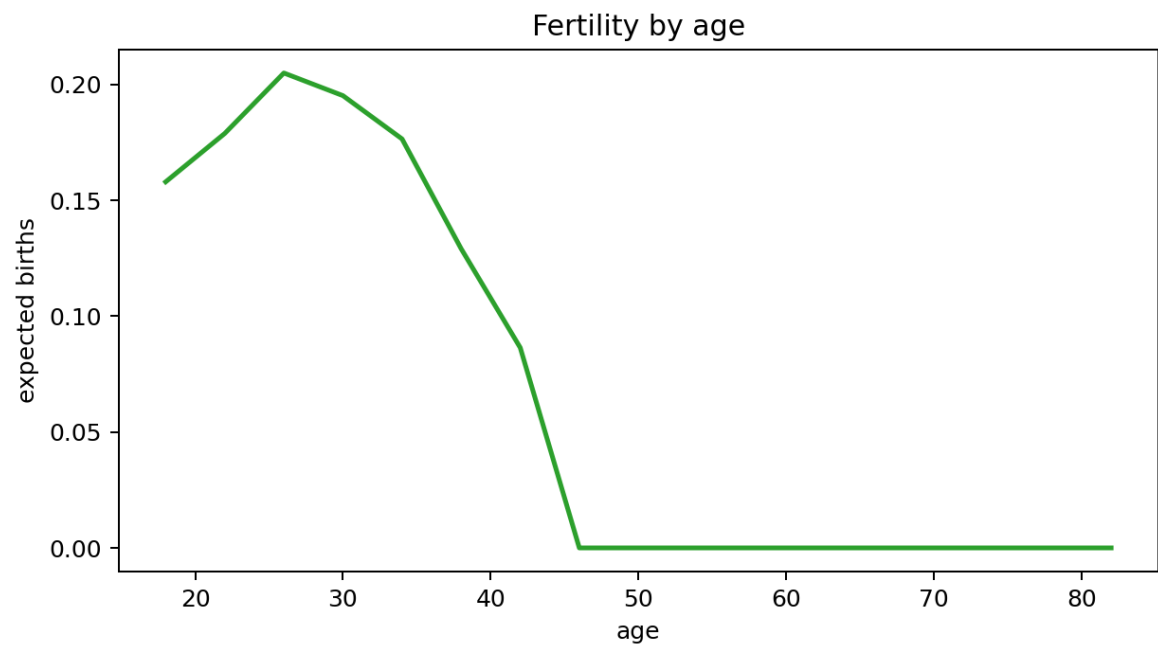
All 17 parameter estimates, bounds, and status

Bounds are taken from the collector's parameters_actual_bounds.csv. 'externally_fixed_or_derived' indicates the parameter is not a searched coordinate in this plan; a missing bound is shown as unavailable.

Parameter	Estimate	Actual lower	Actual upper	Near bound	Status
beta_annual	0.97625	0.94	0.99	False	searched coordinate
kappa_fert	0.20928	0.02	50	True	searched coordinate
kappa_fert_continuation	0.48207	0.02	50	True	searched coordinate
chi	0.88798	0.1	5	False	searched coordinate
H0	8.9672	0.2	80	False	searched coordinate
theta0	0.088169	0	8	False	searched coordinate
theta1	0.087569	0.02	16	True	searched coordinate
first_birth_fixed_cost	0.35327	0	8	False	searched coordinate
h_P	2.0442	0.1	2.3	False	searched coordinate
hbar_child_rooms	0	-	-	-	externally_fixed_or_derived
psi_child	0.14245	-	-	-	externally_fixed_or_derived
payroll_tax	0.179	-	-	-	externally_fixed_or_derived
pension_period	2.0464	-	-	-	externally_fixed_or_derived
housing_supply_elasticity	0.63	-	-	-	externally_fixed_or_derived
tenure_choice_kappa	0.005	-	-	-	externally_fixed_or_derived
alpha_cons	0.733	-	-	-	externally_fixed_or_derived
sigma	2	-	-	-	externally_fixed_or_derived

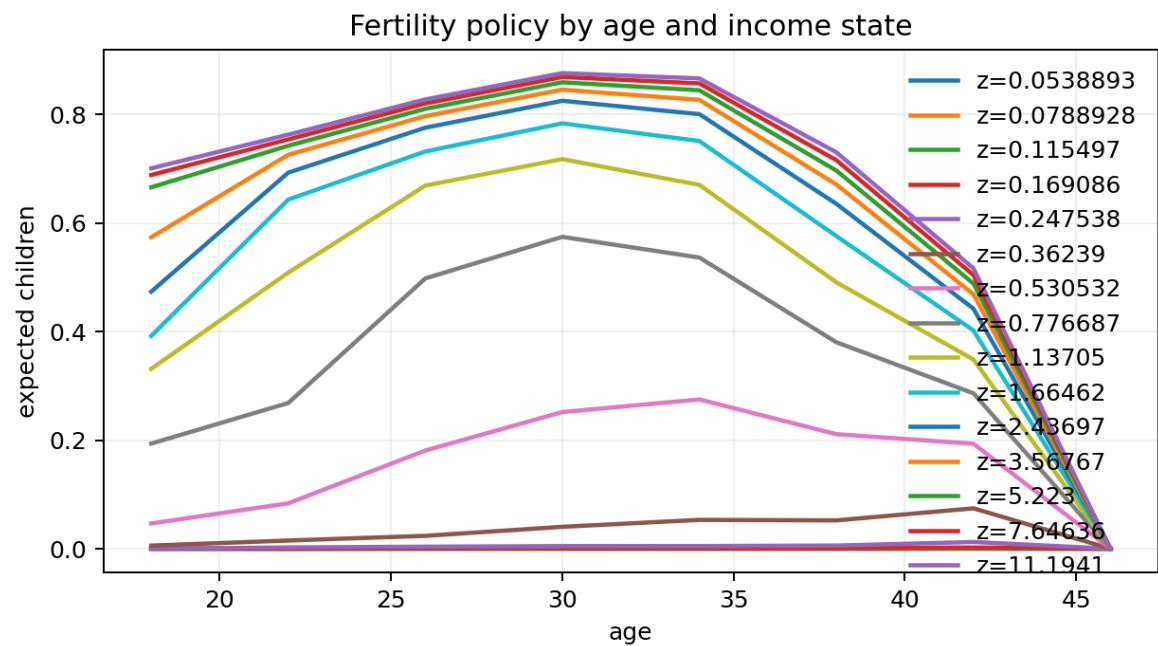
Appendix: B_floor original standard diagnostics (1-2 of 17)

fertility by age



Original diagnostic image retained unchanged.

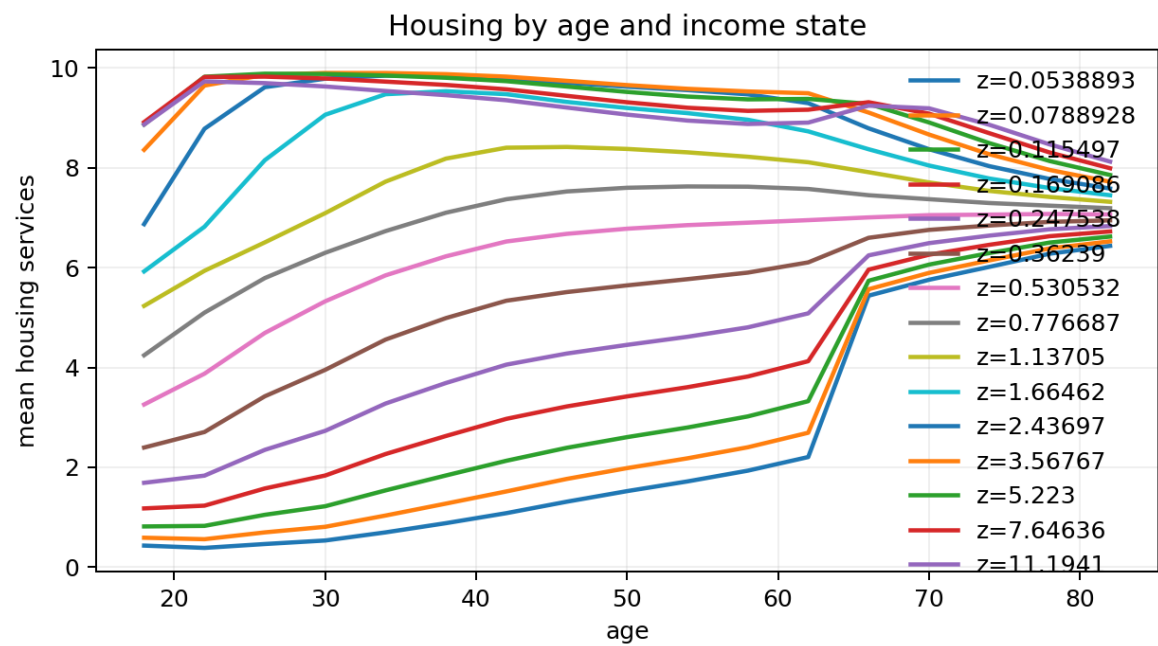
fertility policy by age income state



Original diagnostic image retained unchanged.

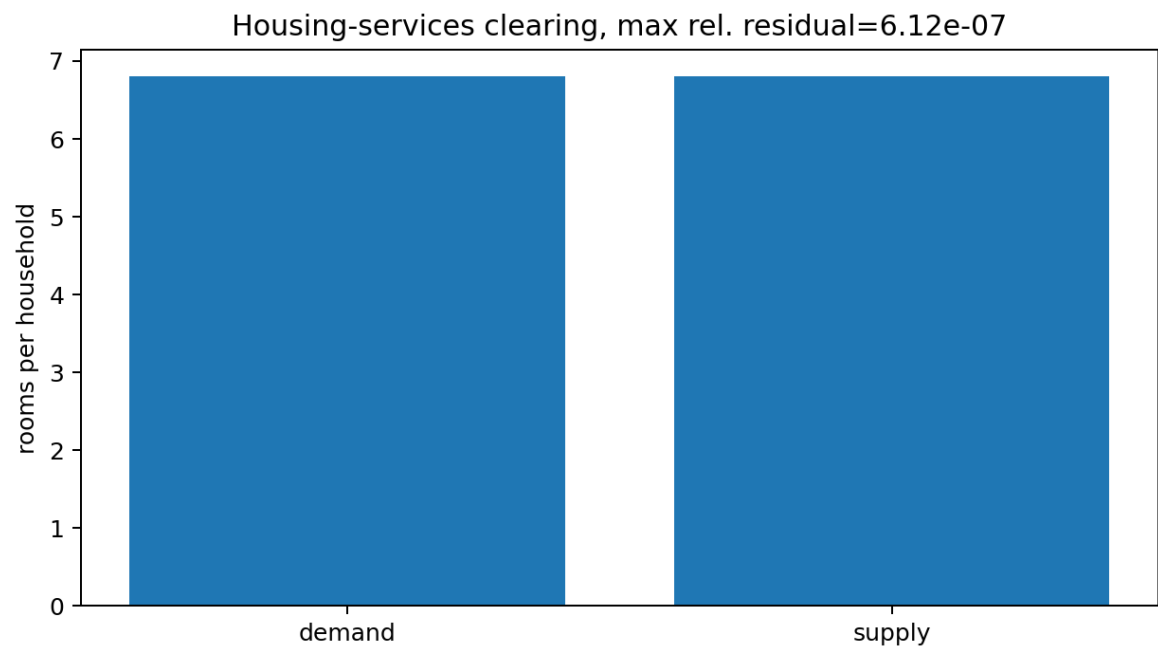
Appendix: B_floor original standard diagnostics (3-4 of 17)

housing by age income state



Original diagnostic image retained unchanged.

housing market



Original diagnostic image retained unchanged.

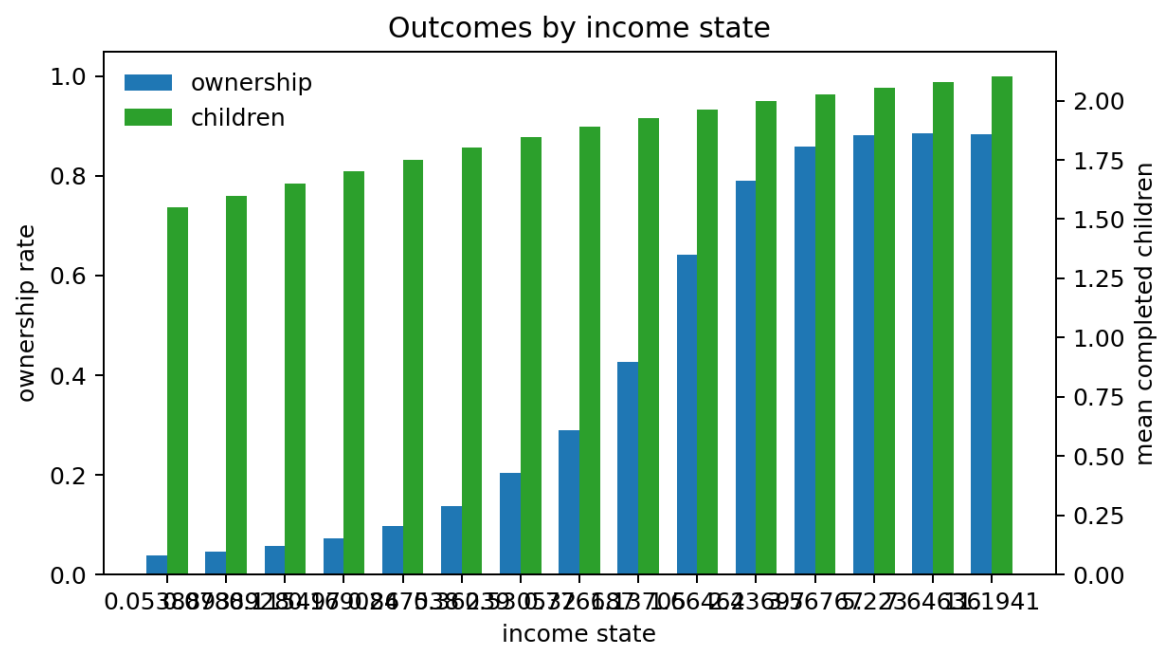
Appendix: B_floor original standard diagnostics (5-6 of 17)

housing prices



Original diagnostic image retained unchanged.

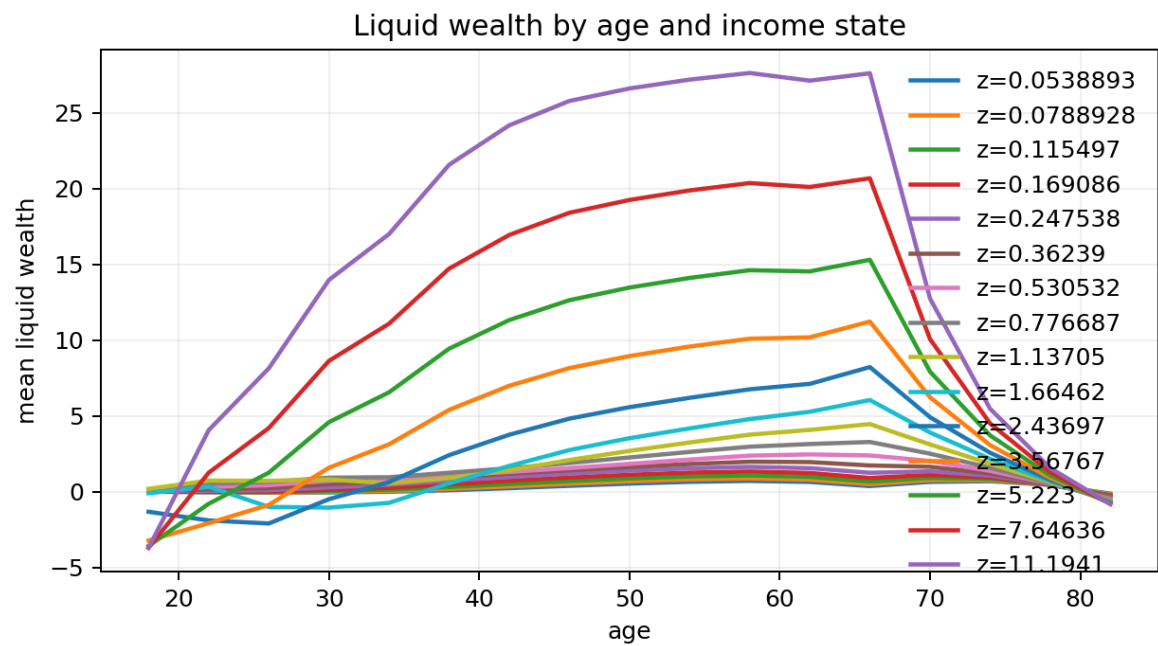
income state outcomes



Original diagnostic image retained unchanged.

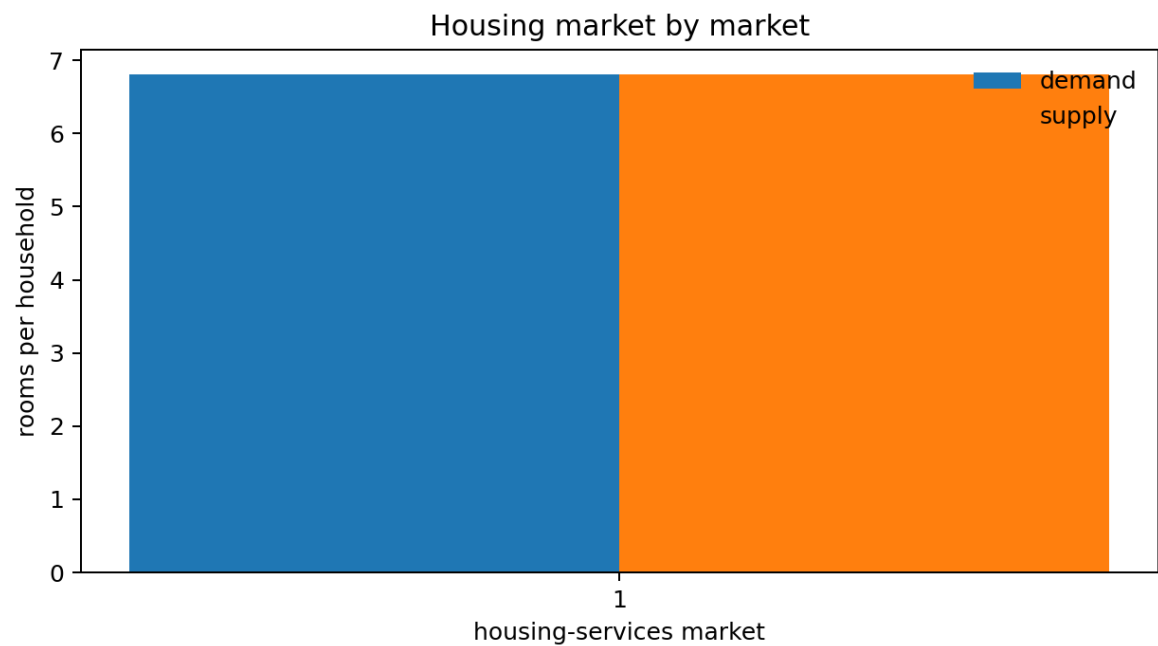
Appendix: B_floor original standard diagnostics (7-8 of 17)

liquid wealth by age income state



Original diagnostic image retained unchanged.

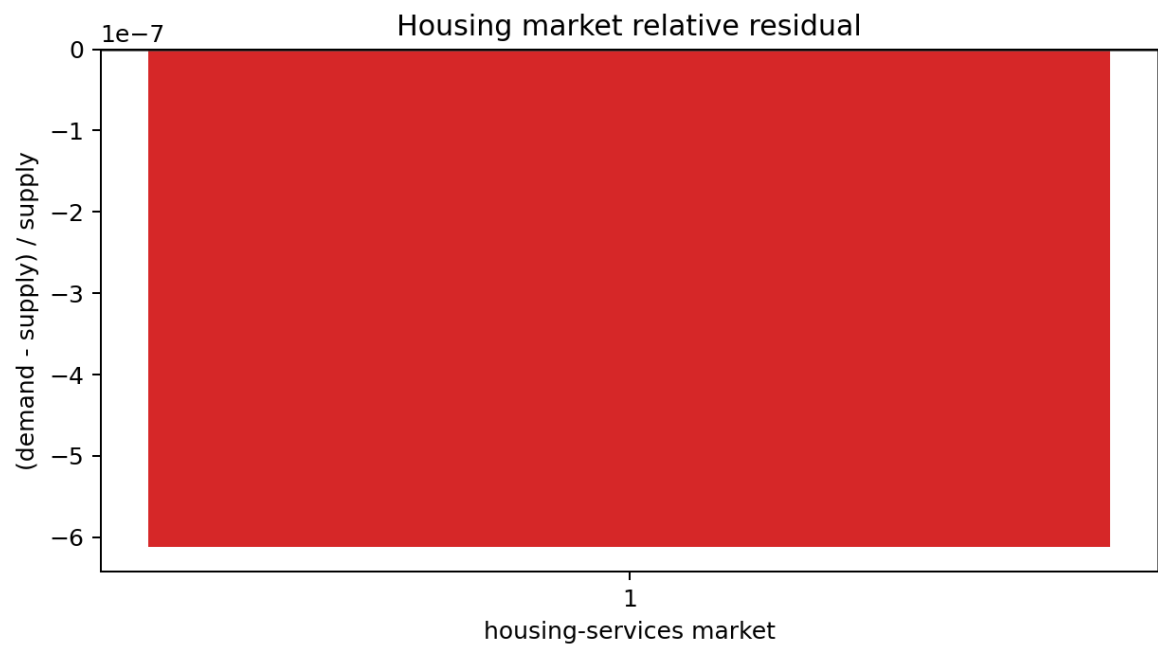
market clearing by market



Original diagnostic image retained unchanged.

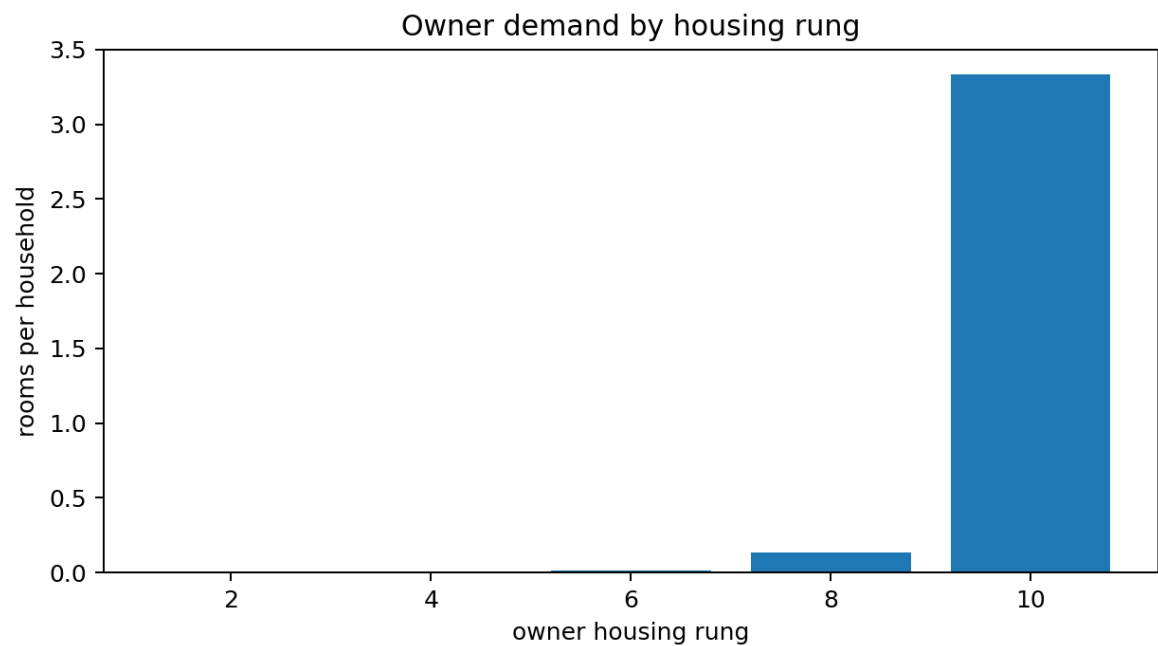
Appendix: B_floor original standard diagnostics (9-10 of 17)

market clearing residuals



Original diagnostic image retained unchanged.

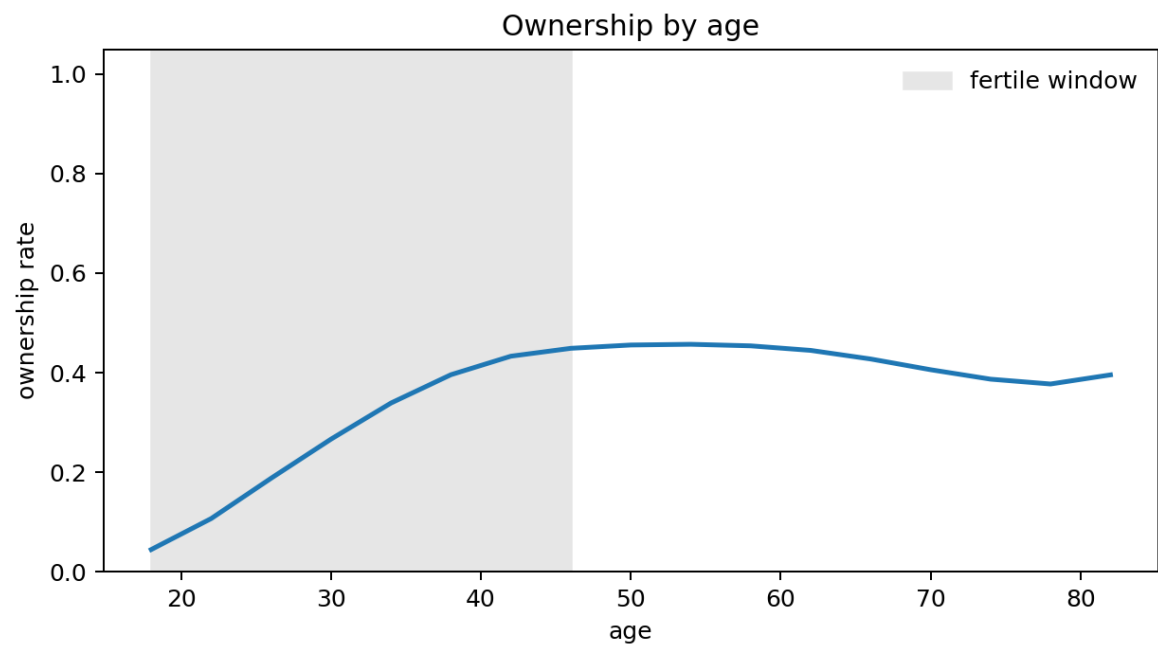
owner rungs



Original diagnostic image retained unchanged.

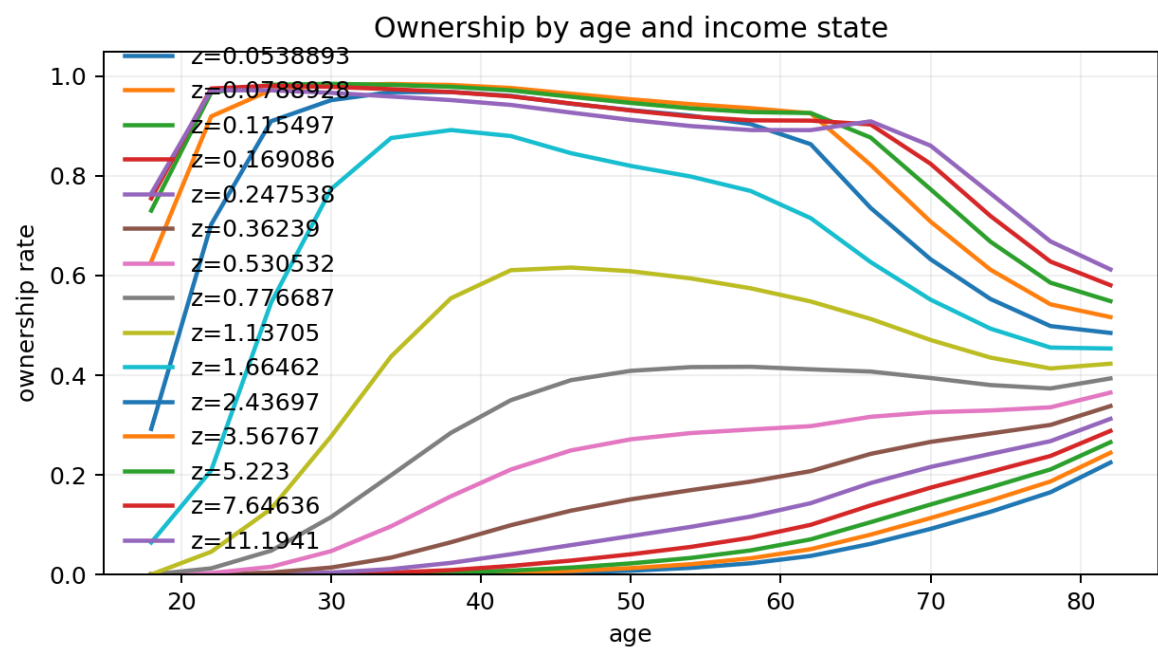
Appendix: B_floor original standard diagnostics (11-12 of 17)

ownership by age



Original diagnostic image retained unchanged.

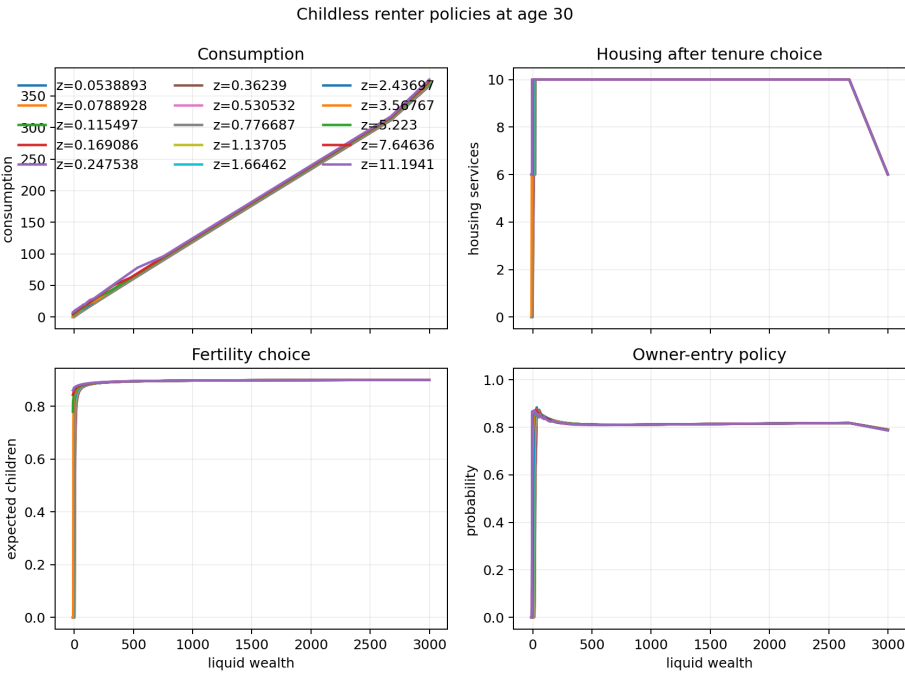
ownership by age income state



Original diagnostic image retained unchanged.

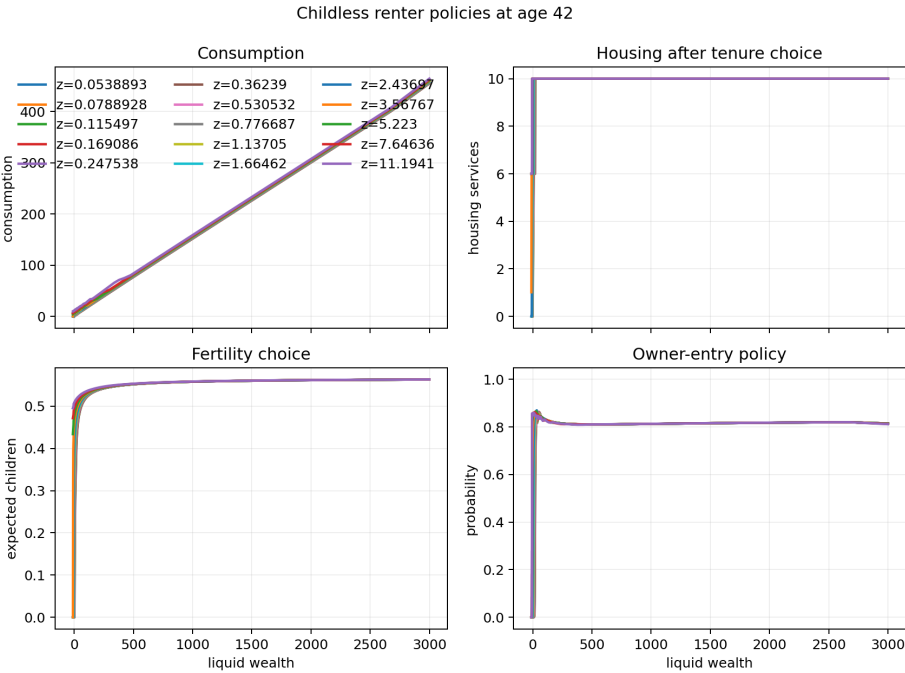
Appendix: B_floor original standard diagnostics (13-14 of 17)

policy childless renter age30



Original diagnostic image retained unchanged.

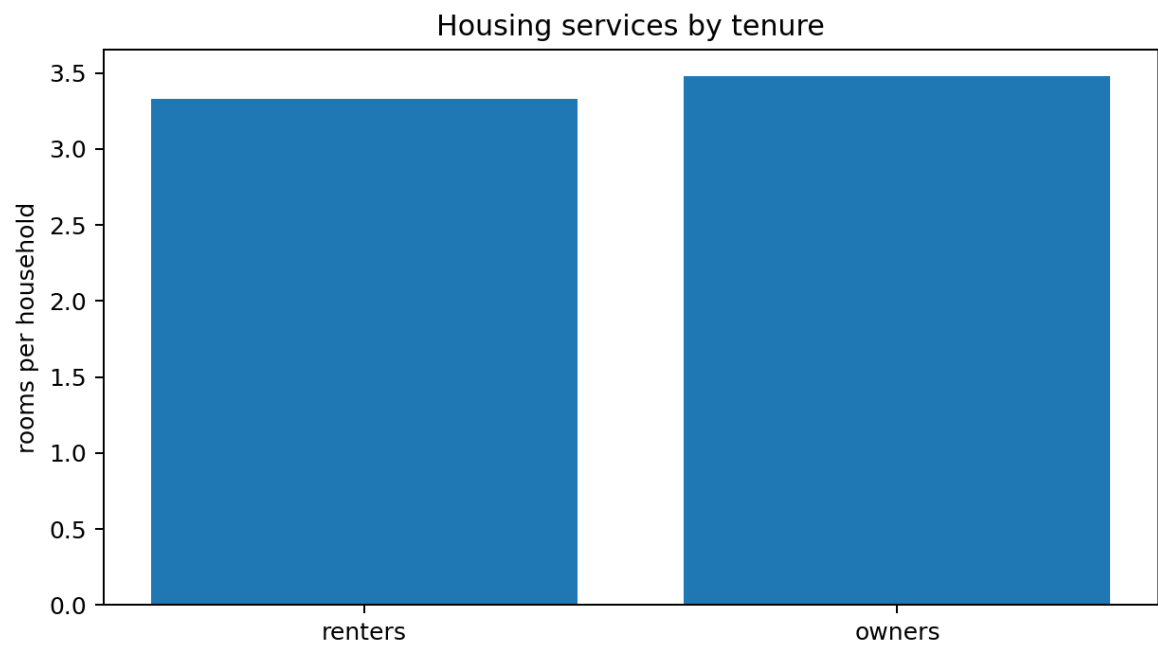
policy childless renter age42



Original diagnostic image retained unchanged.

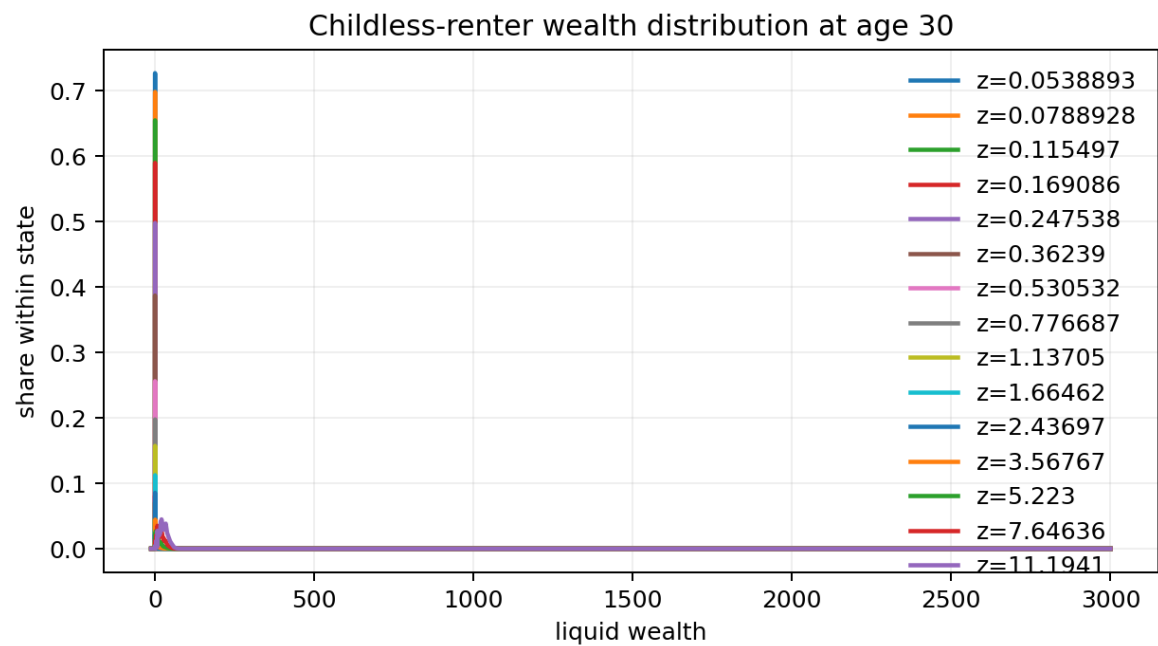
Appendix: B_floor original standard diagnostics (15-16 of 17)

tenure services



Original diagnostic image retained unchanged.

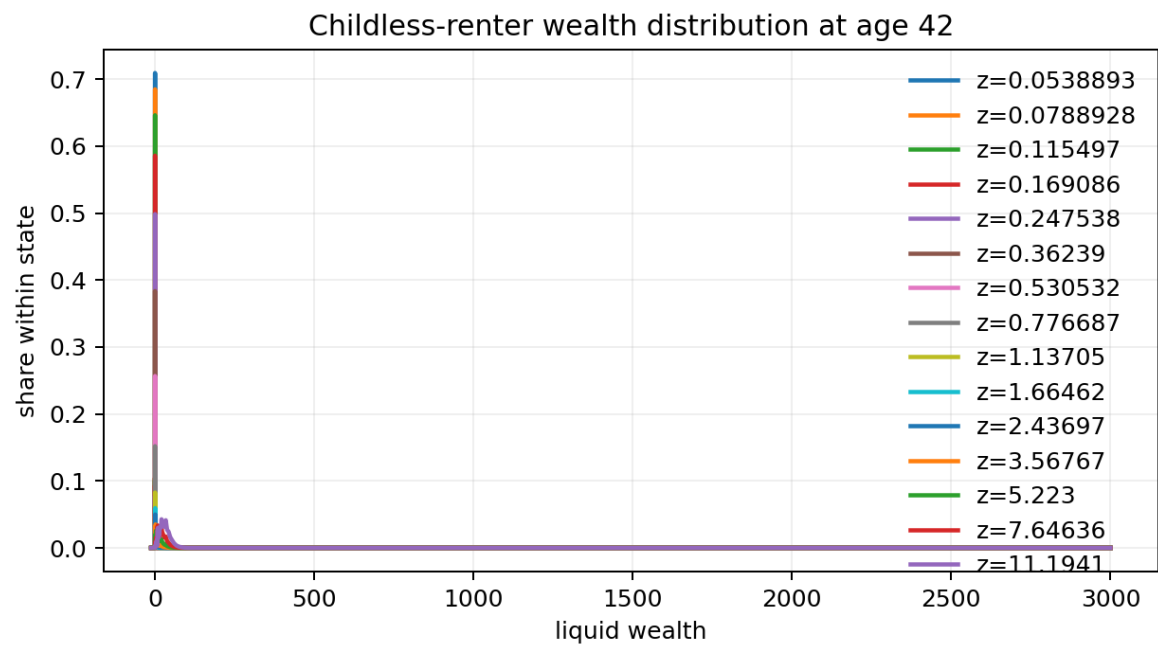
wealth dist childless renter age30



Original diagnostic image retained unchanged.

Appendix: B_floor original standard diagnostics (17-17 of 17)

wealth dist childless renter age42



Original diagnostic image retained unchanged.

Appendix: B_shares full selected target fit

Collector-selected stage: production; case: worker33_proposal01; loss: 851.271338. Collection verification receipt checkpoint SHA-256: 5c382f34f852b10efc2d4253753ce8811c504b3cf95c4d4cce89399e0c60a5f4.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Initial fertility normalization	2.1	2.1	2.3775e-05	-	-
Childless women, ages 40-44	0.19828	0.1496	-0.048679	35,532.30	84.198
Exactly one child among mothers, ages 40-44	0.21366	0.25282	0.039169	26,952.82	41.351
Period mean first-birth age	25.976	25.727	-0.2493	139.83	8.6901
First births at age 30+	0.24928	0.21398	-0.035293	13,866.07	17.272
Wealth / annual gross labor earnings	6.1459	6.8694	0.72351	7.5951	3.9758
Annual bequests / aggregate wealth	0.0088	0.0069287	-0.0018713	5,165,289.26	18.088
Old wealth/income p90 / median, ages 76-84	3.5159	3.1184	-0.39751	10.616	1.6776
Mean occupied rooms, capped at 9	5.5611	7.0882	1.5271	128.02	298.55
Ownership, heads 30-55	0.64833	0.51325	-0.13508	2,339.36	42.688
First-birth room response, -1 to +3	0.72025	1.0437	0.32341	137.57	14.388
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.49163	0.14456	280.53	5.8626
Recent-parent ownership gap	0.1629	0.055075	-0.10782	27,055.82	314.53

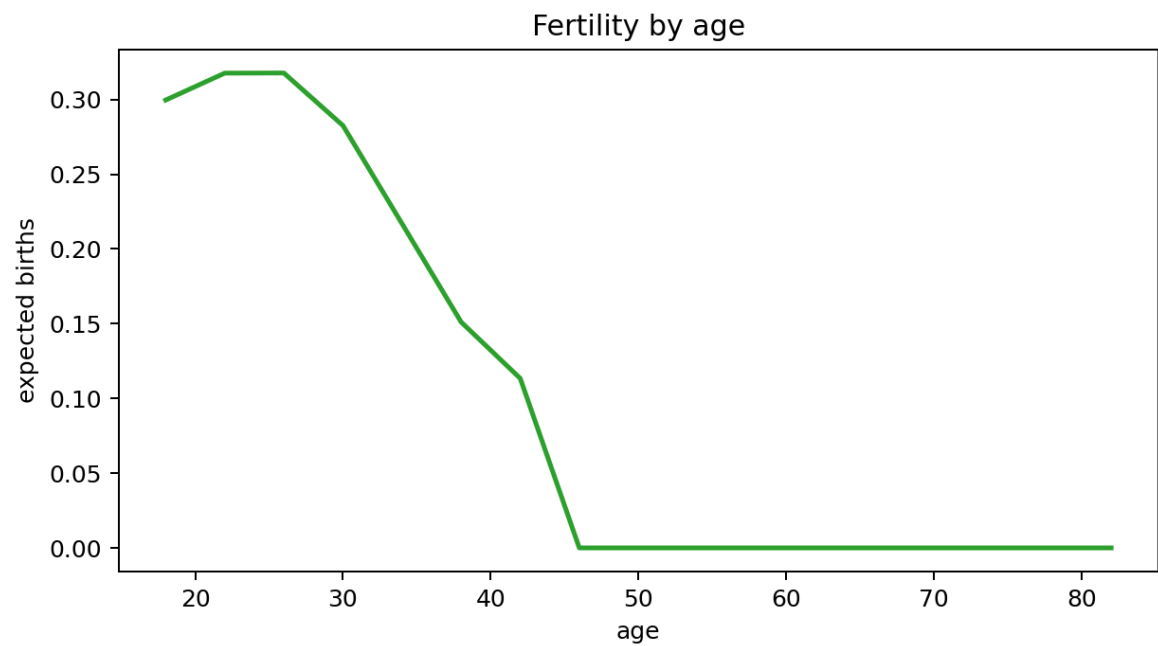
All 19 parameter estimates, bounds, and status

Bounds are taken from the collector's parameters_actual_bounds.csv. 'externally_fixed_or_derived' indicates the parameter is not a searched coordinate in this plan; a missing bound is shown as unavailable.

Parameter	Estimate	Actual lower	Actual upper	Near bound	Status
beta_annual	0.99	0.94	0.99	True	searched coordinate
kappa_fert	0.22235	0.02	50	True	searched coordinate
kappa_fert_continuation	0.29376	0.02	50	True	searched coordinate
chi	0.93878	0.1	5	False	searched coordinate
H0	10.576	0.2	80	False	searched coordinate
theta0	0.074686	0	8	True	searched coordinate
theta1	0.099049	0.02	16	True	searched coordinate
first_birth_fixed_cost	0.55905	0	8	False	searched coordinate
delta_alpha_jump	0.051345	0	0.25	False	searched coordinate
delta_alpha	0.032877	0	0.25	False	searched coordinate
hbar_first_child_jump	0	-	-	-	externally_fixed_or_derived
hbar_child_rooms	0	-	-	-	externally_fixed_or_derived
psi_child	0.04492	-	-	-	externally_fixed_or_derived
payroll_tax	0.179	-	-	-	externally_fixed_or_derived
pension_period	2.0464	-	-	-	externally_fixed_or_derived
housing_supply_elasticity	0.63	-	-	-	externally_fixed_or_derived
tenure_choice_kappa	0.005	-	-	-	externally_fixed_or_derived
alpha_cons	0.733	-	-	-	externally_fixed_or_derived
sigma	2	-	-	-	externally_fixed_or_derived

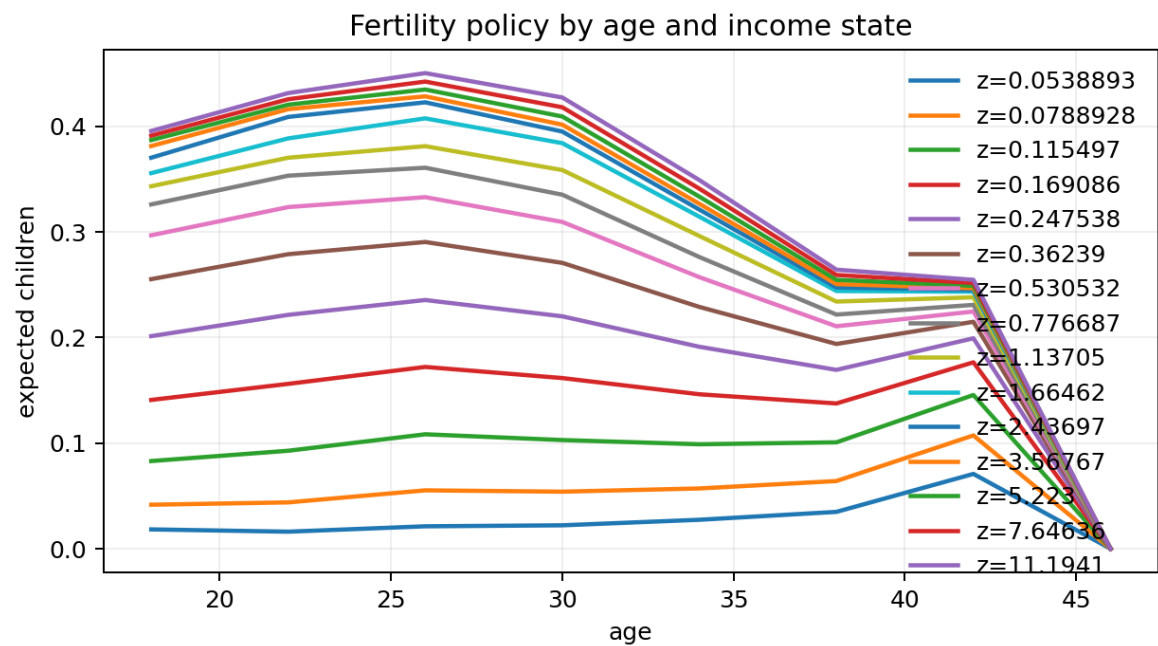
Appendix: B_shares original standard diagnostics (1-2 of 17)

fertility by age



Original diagnostic image retained unchanged.

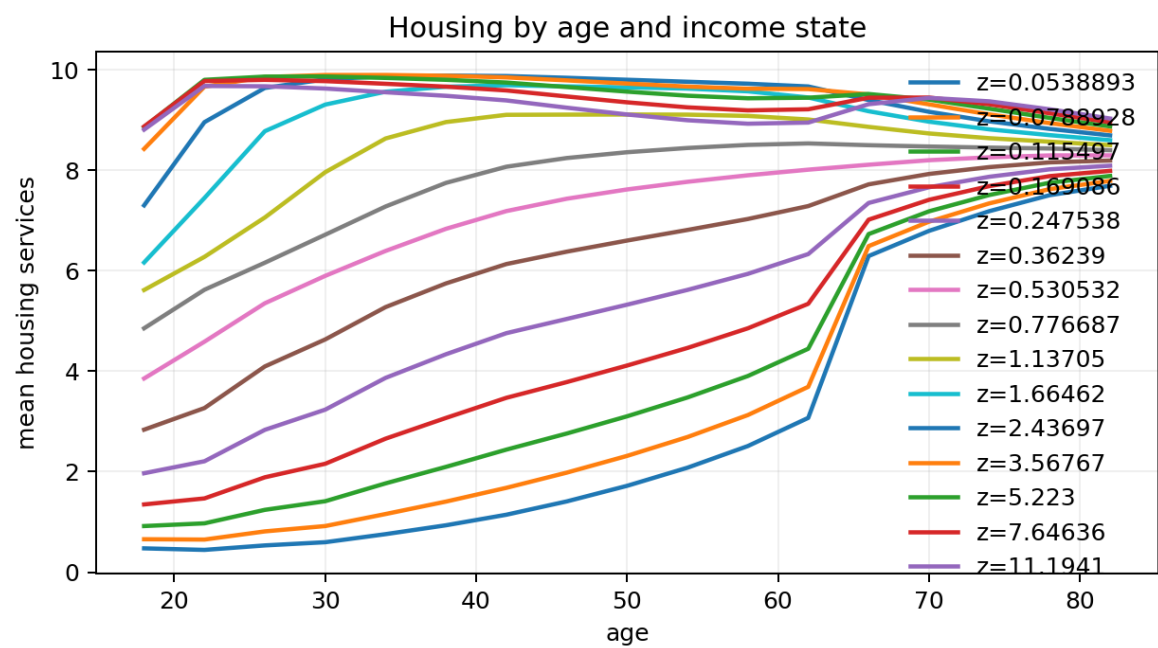
fertility policy by age income state



Original diagnostic image retained unchanged.

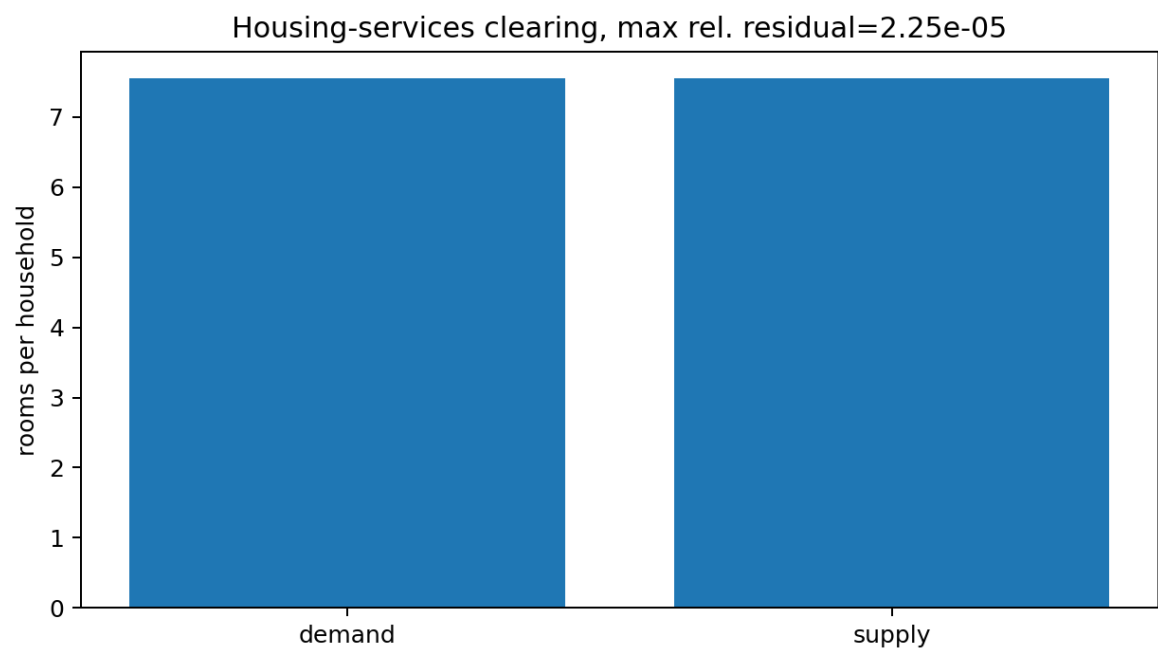
Appendix: B_shares original standard diagnostics (3-4 of 17)

housing by age income state



Original diagnostic image retained unchanged.

housing market



Original diagnostic image retained unchanged.

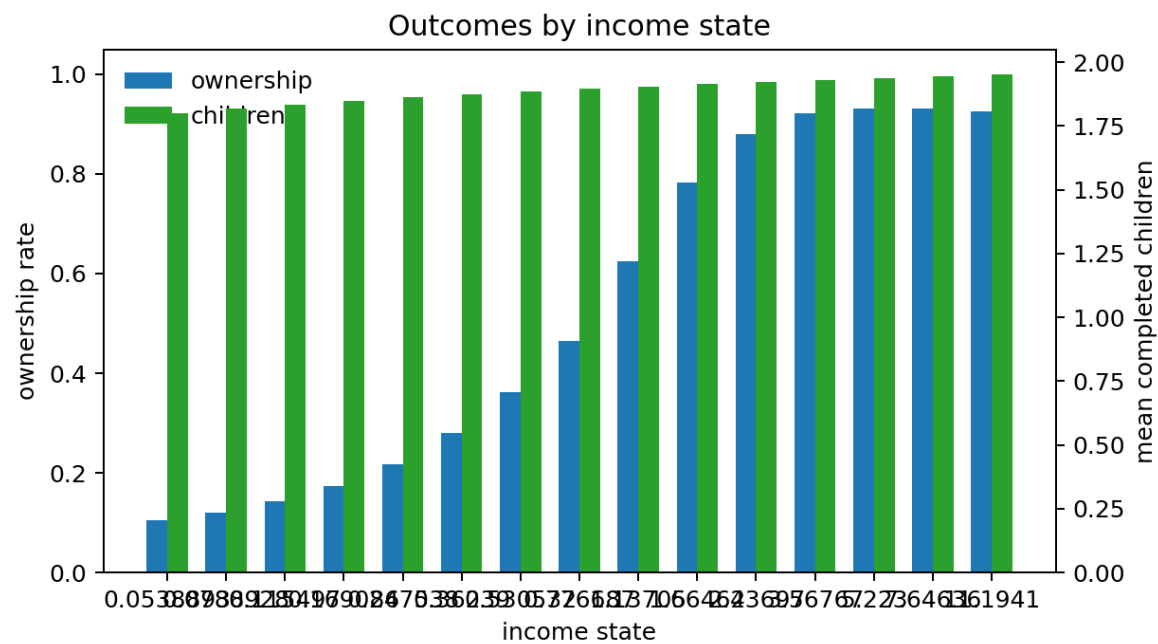
Appendix: B_shares original standard diagnostics (5-6 of 17)

housing prices



Original diagnostic image retained unchanged.

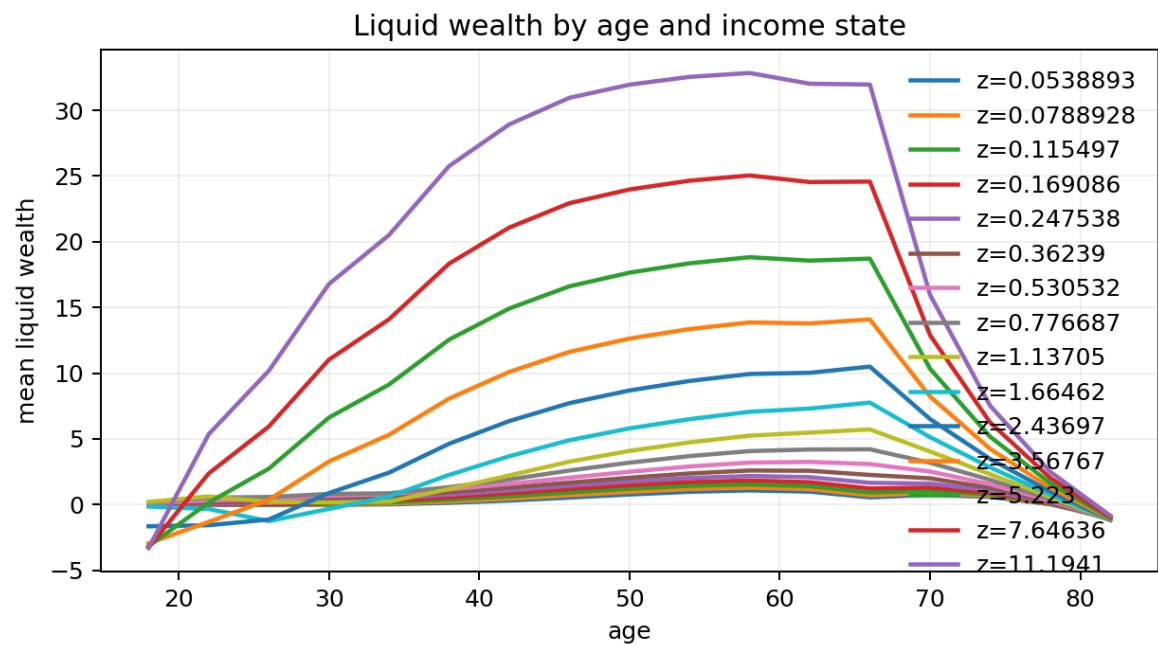
income state outcomes



Original diagnostic image retained unchanged.

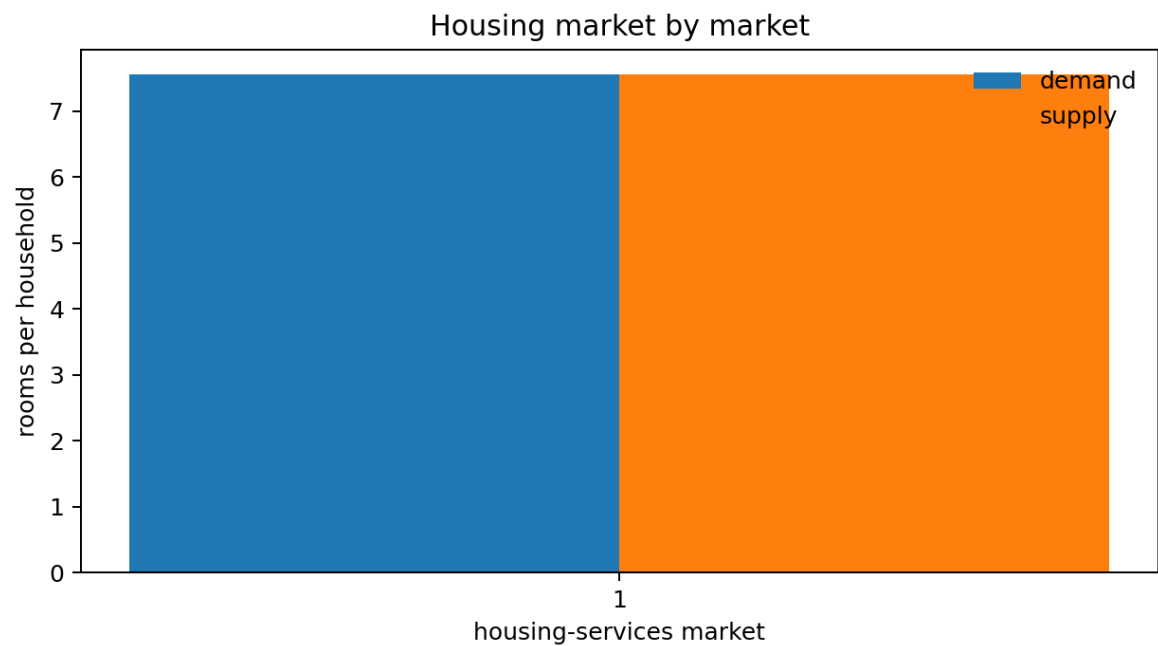
Appendix: B_shares original standard diagnostics (7-8 of 17)

liquid wealth by age income state



Original diagnostic image retained unchanged.

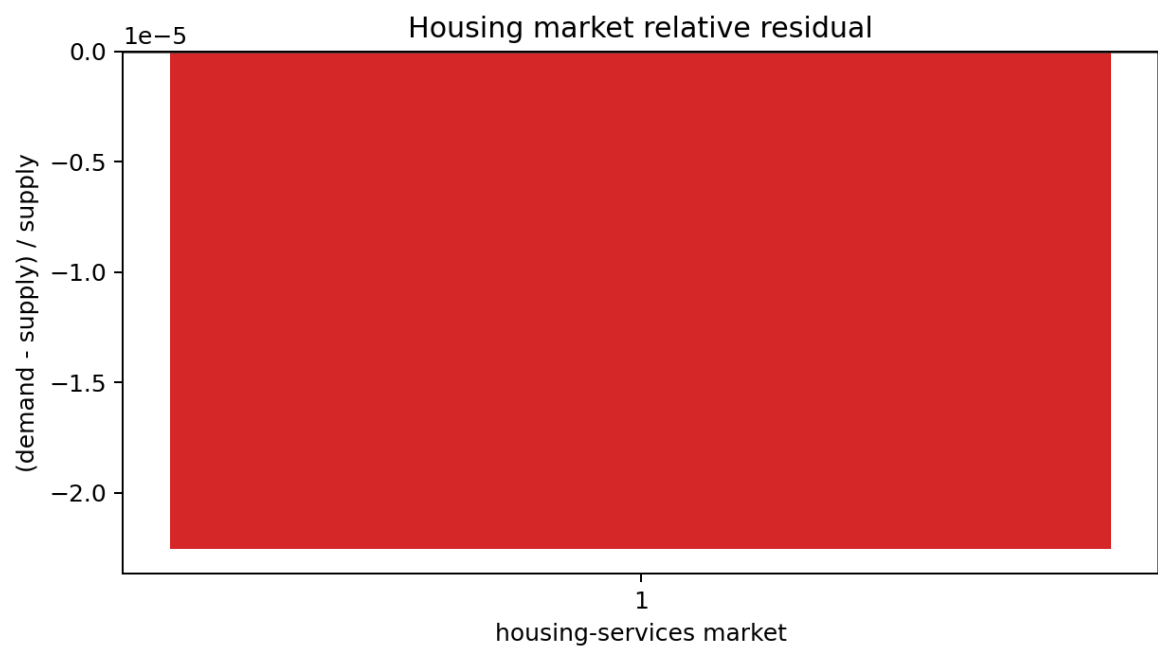
market clearing by market



Original diagnostic image retained unchanged.

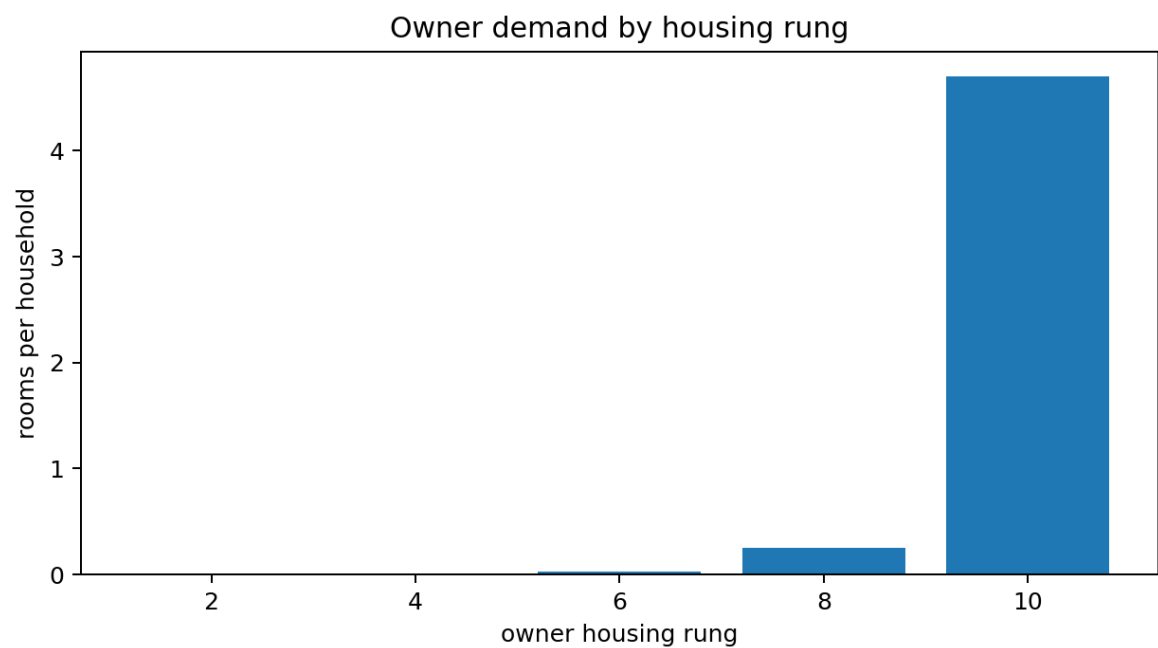
Appendix: B_shares original standard diagnostics (9-10 of 17)

market clearing residuals



Original diagnostic image retained unchanged.

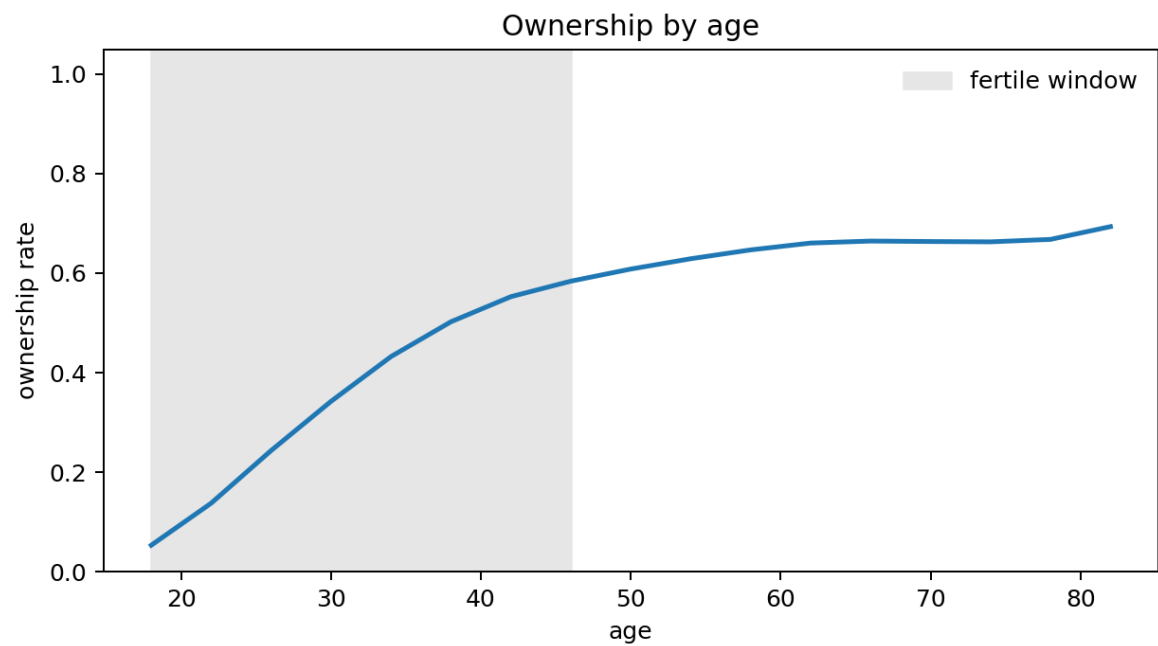
owner rungs



Original diagnostic image retained unchanged.

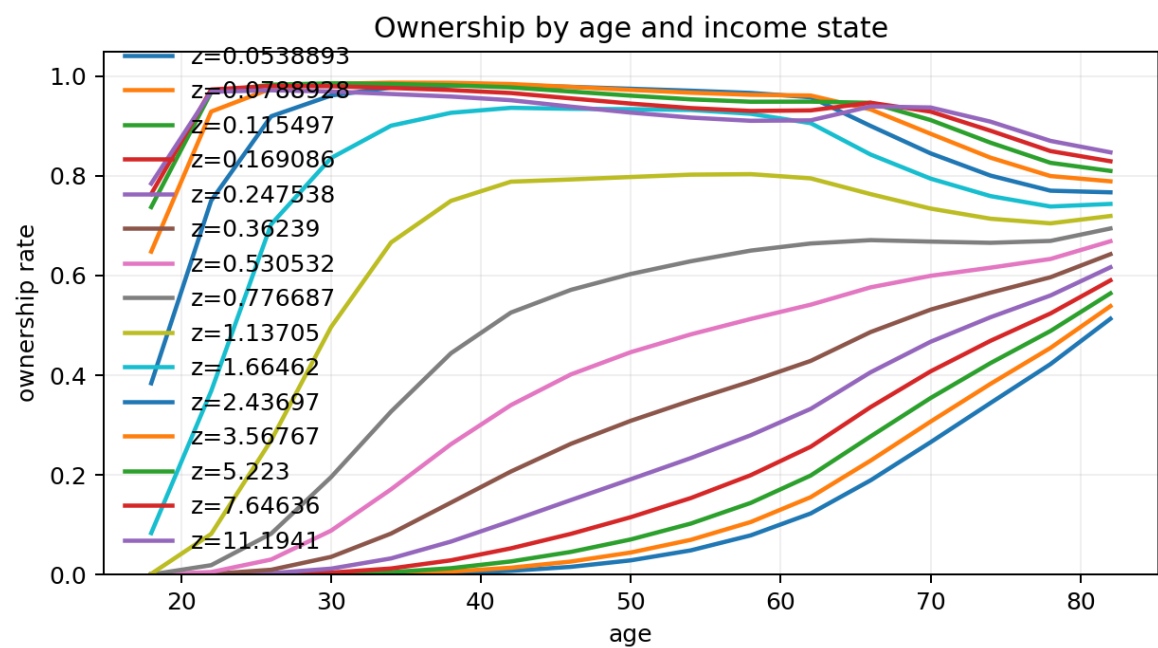
Appendix: B_shares original standard diagnostics (11-12 of 17)

ownership by age



Original diagnostic image retained unchanged.

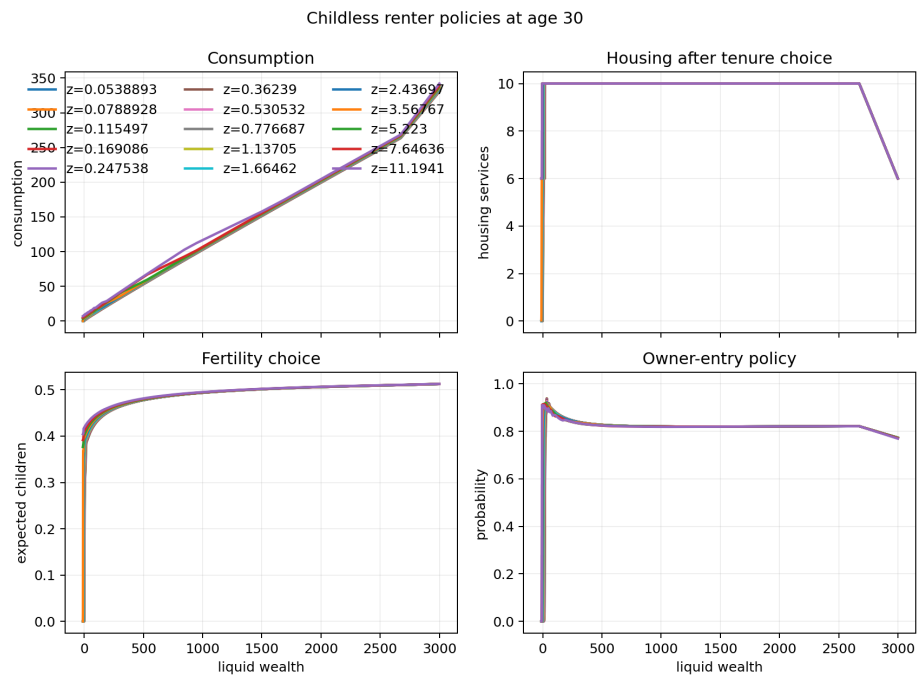
ownership by age income state



Original diagnostic image retained unchanged.

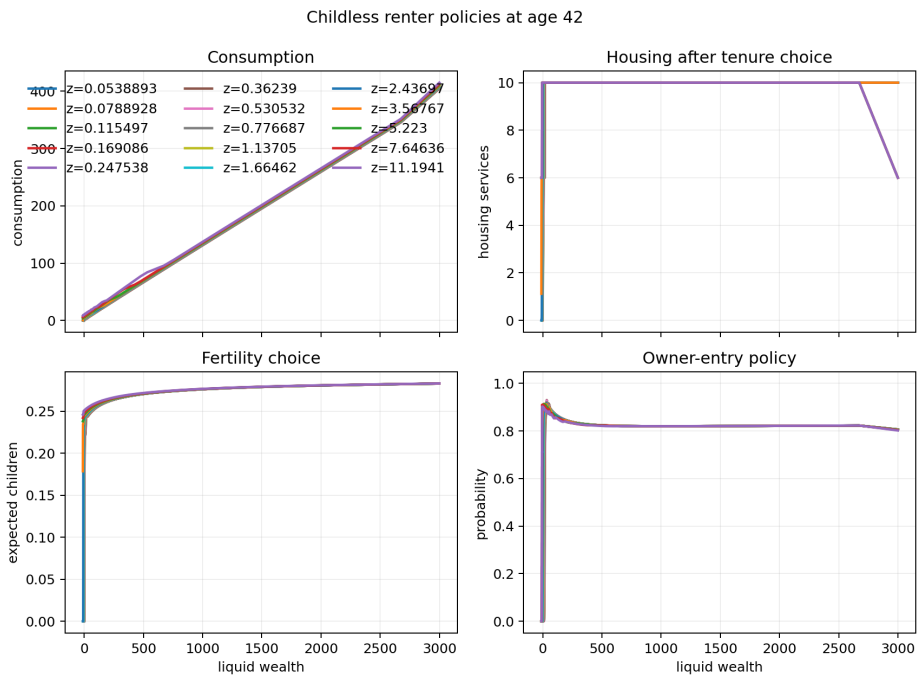
Appendix: B_shares original standard diagnostics (13-14 of 17)

policy childless renter age30



Original diagnostic image retained unchanged.

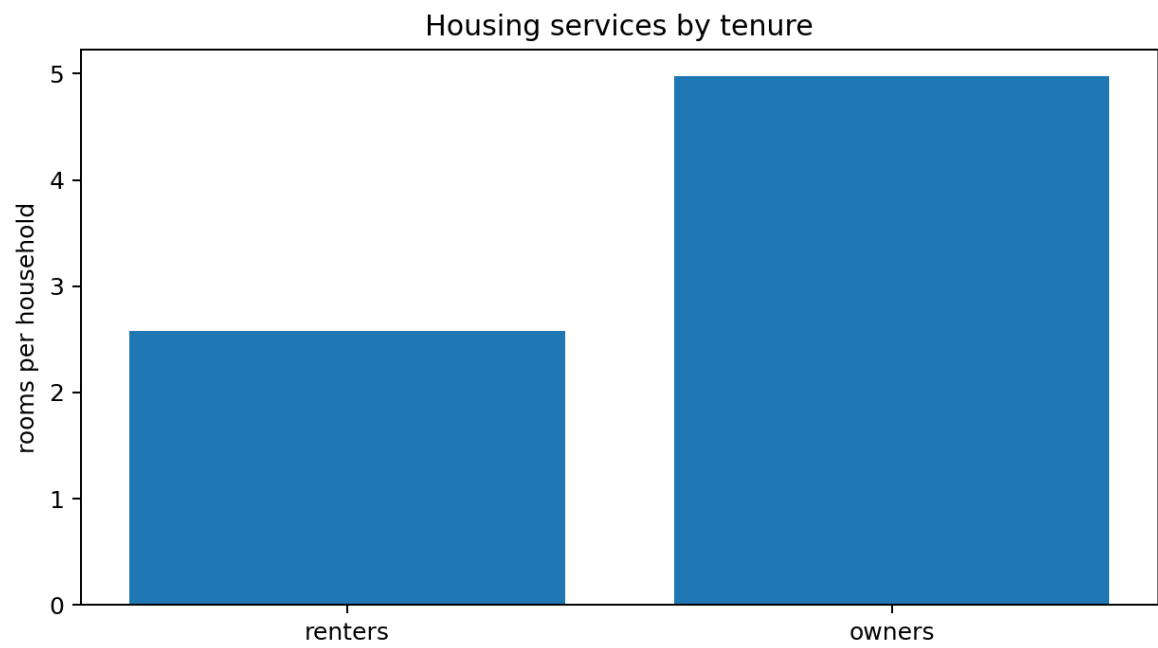
policy childless renter age42



Original diagnostic image retained unchanged.

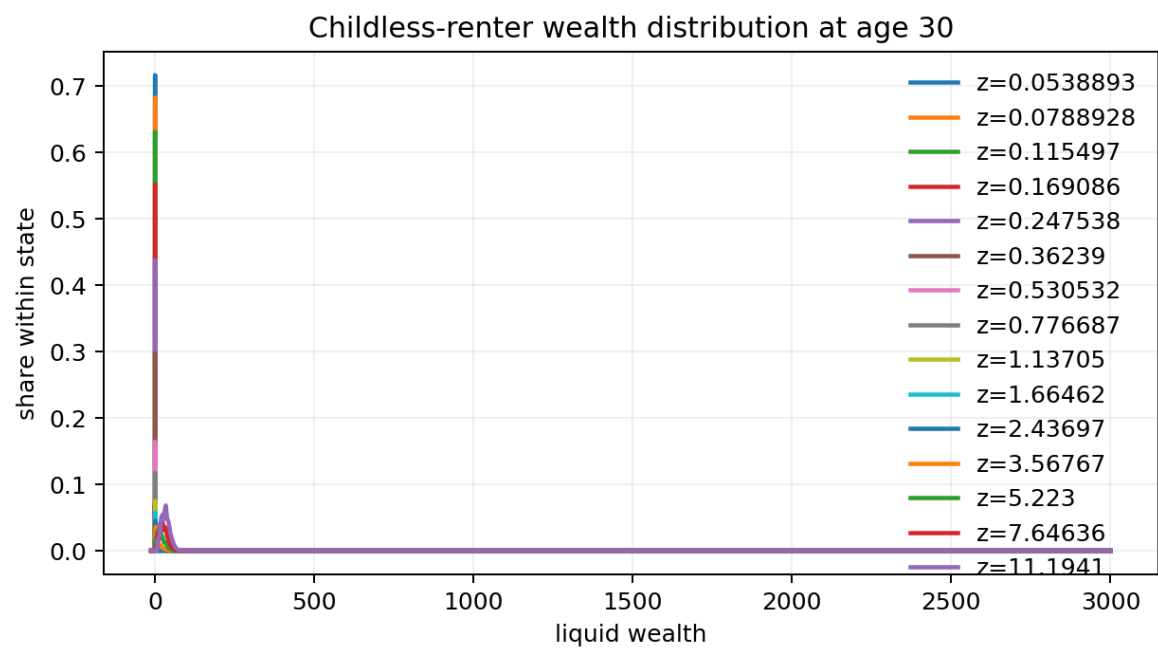
Appendix: B_shares original standard diagnostics (15-16 of 17)

tenure services



Original diagnostic image retained unchanged.

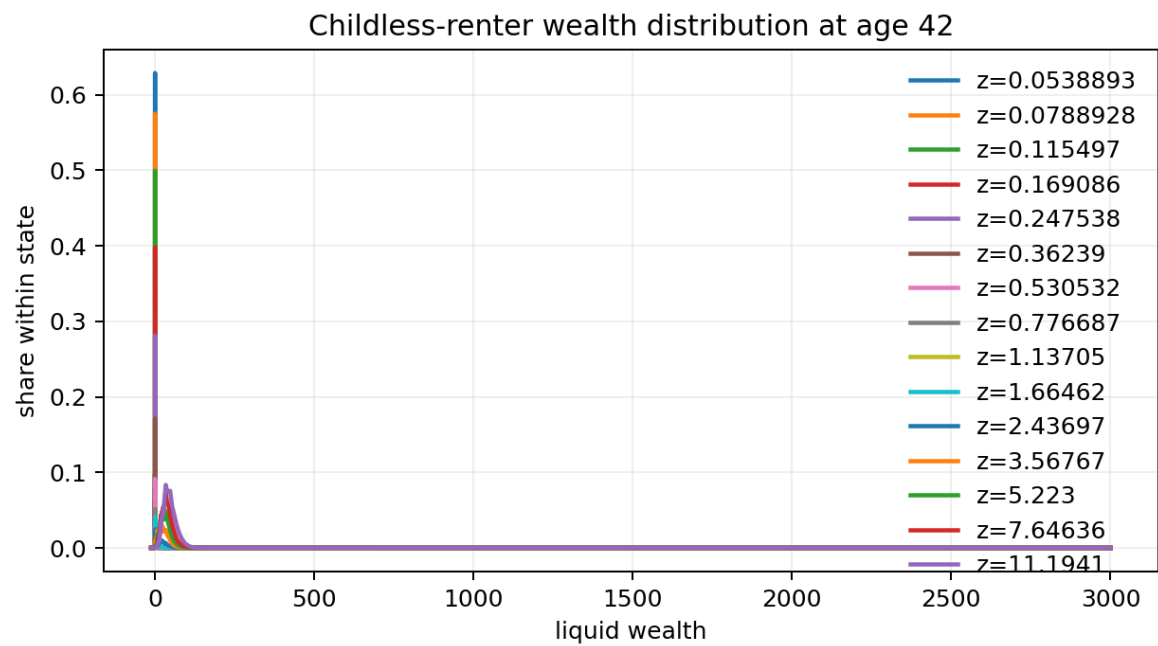
wealth dist childless renter age30



Original diagnostic image retained unchanged.

Appendix: B_shares original standard diagnostics (17-17 of 17)

wealth dist childless renter age42



Original diagnostic image retained unchanged.

Appendix: D_floor full selected target fit

Collector-selected stage: production; case: worker37_proposal17; loss: 416.481444. Collection verification receipt checkpoint SHA-256: e4afbe7ca2197c5ea8f2dbde2ccaf0bdde9a286932818c2466eb186f48b467ae.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Initial fertility normalization	2.1	2.1001	8.6678e-05	-	-
Childless women, ages 40-44	0.19828	0.1628	-0.035481	35,532.30	44.732
Exactly one child among mothers, ages 40-44	0.21366	0.25062	0.036969	26,952.82	36.837
Period mean first-birth age	25.976	26.409	0.43231	139.83	26.133
First births at age 30+	0.24928	0.24933	5.6812e-05	13,866.07	4.4755e-05
Wealth / annual gross labor earnings	6.1459	5.4899	-0.65598	7.5951	3.2683
Annual bequests / aggregate wealth	0.0088	0.0061056	-0.0026944	5,165,289.26	37.499
Old wealth/income p90 / median, ages 76-84	3.5159	4.3821	0.8662	10.616	7.9656
Mean occupied rooms, capped at 9	5.5611	6.5219	0.96081	128.02	118.18
Ownership, heads 30-55	0.64833	0.44551	-0.20283	2,339.36	96.237
First-birth room response, -1 to +3	0.72025	1.1473	0.42702	137.57	25.084
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.29649	-0.050576	280.53	0.71757
Recent-parent ownership gap	0.1629	0.13583	-0.027068	27,055.82	19.824

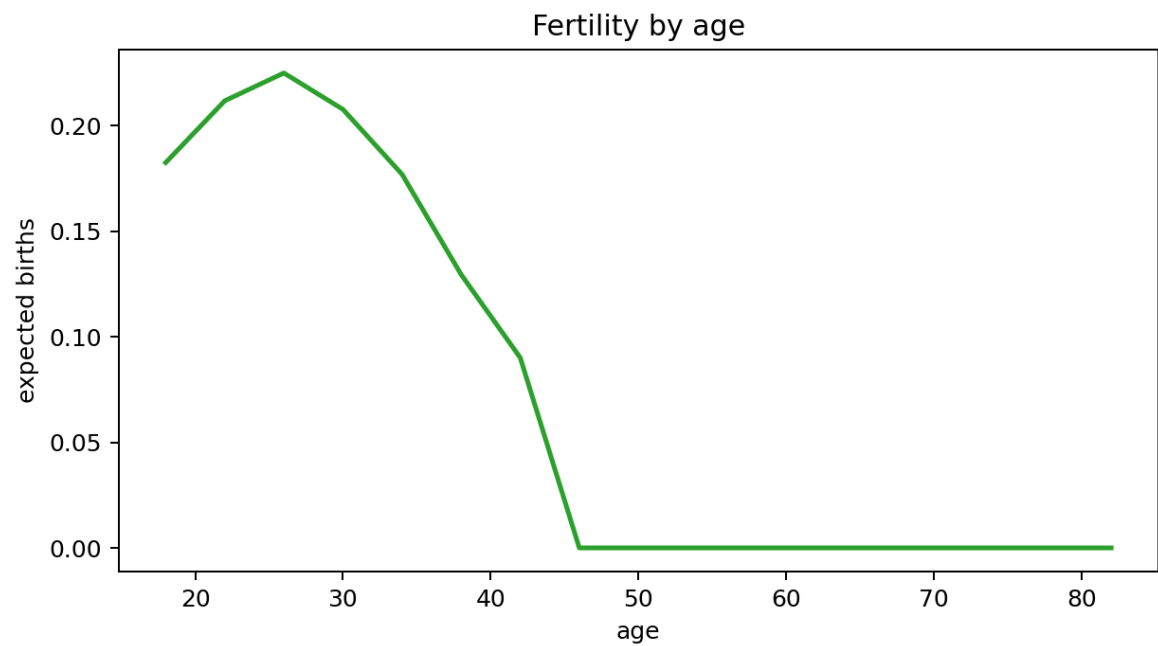
All 17 parameter estimates, bounds, and status

Bounds are taken from the collector's parameters_actual_bounds.csv. 'externally_fixed_or_derived' indicates the parameter is not a searched coordinate in this plan; a missing bound is shown as unavailable.

Parameter	Estimate	Actual lower	Actual upper	Near bound	Status
beta_annual	0.98241	0.94	0.99	False	searched coordinate
kappa_fert	0.22842	0.02	50	True	searched coordinate
kappa_fert_continuation	0.48531	0.02	50	True	searched coordinate
chi	0.95607	0.1	5	False	searched coordinate
H0	8.749	0.2	80	False	searched coordinate
theta0	0.033355	0	8	True	searched coordinate
theta1	0.091563	0.02	16	True	searched coordinate
first_birth_fixed_cost	0.33446	0	8	False	searched coordinate
h_P	2.1493	0.1	2.3	False	searched coordinate
hbar_child_rooms	0	-	-	-	externally_fixed_or_derived
psi_child	0.13707	-	-	-	externally_fixed_or_derived
payroll_tax	0.179	-	-	-	externally_fixed_or_derived
pension_period	2.0464	-	-	-	externally_fixed_or_derived
housing_supply_elasticity	0.63	-	-	-	externally_fixed_or_derived
tenure_choice_kappa	0.005	-	-	-	externally_fixed_or_derived
alpha_cons	0.733	-	-	-	externally_fixed_or_derived
sigma	2	-	-	-	externally_fixed_or_derived

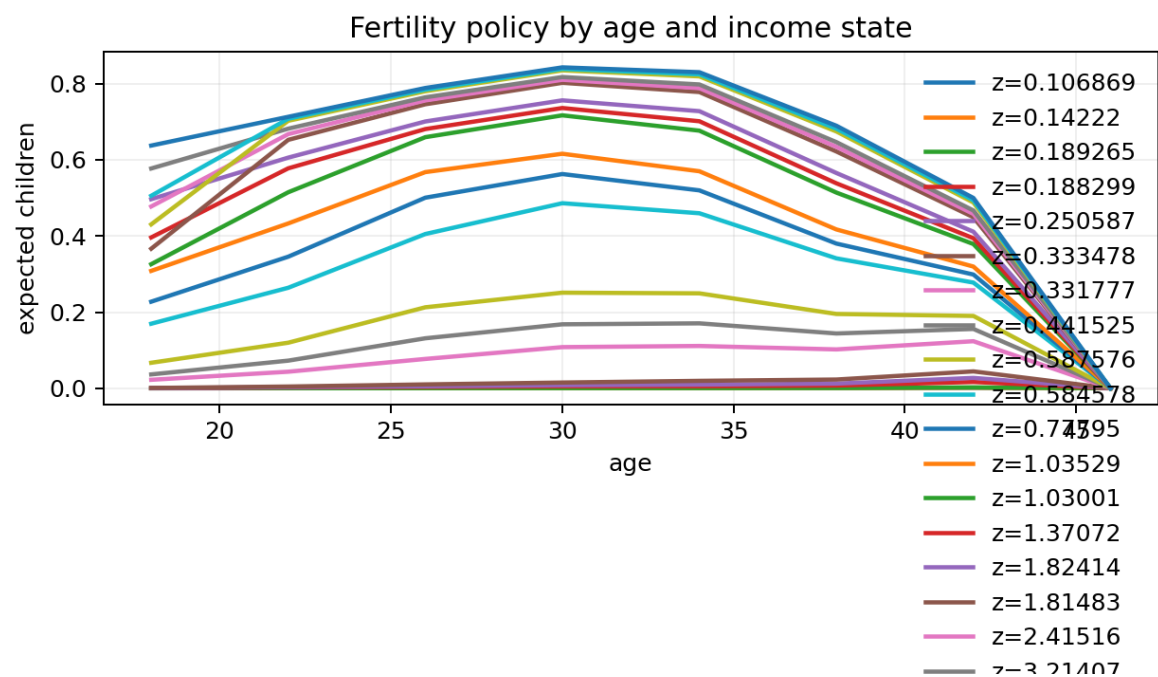
Appendix: D_floor original standard diagnostics (1-2 of 17)

fertility by age



Original diagnostic image retained unchanged.

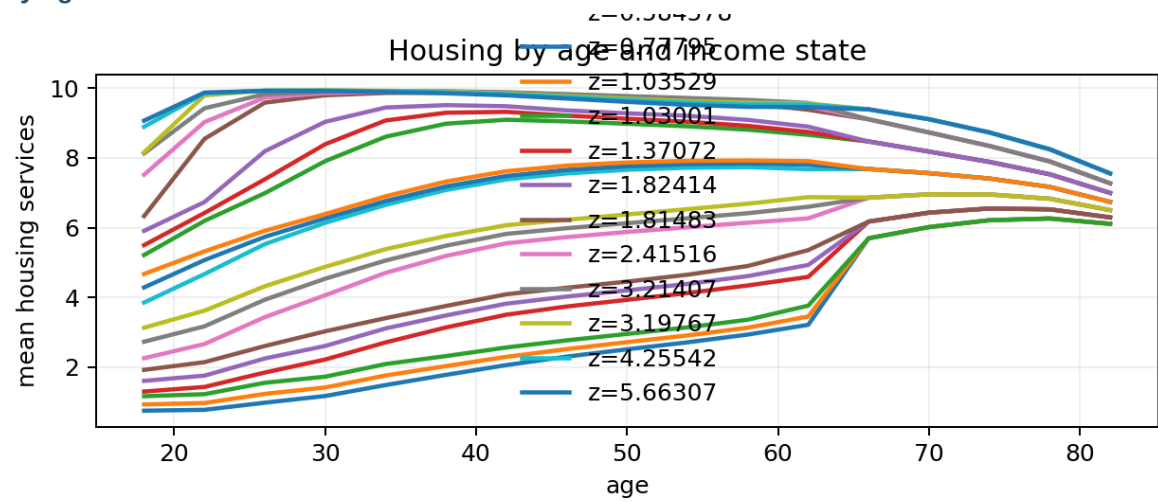
fertility policy by age income state



Original diagnostic image retained unchanged.

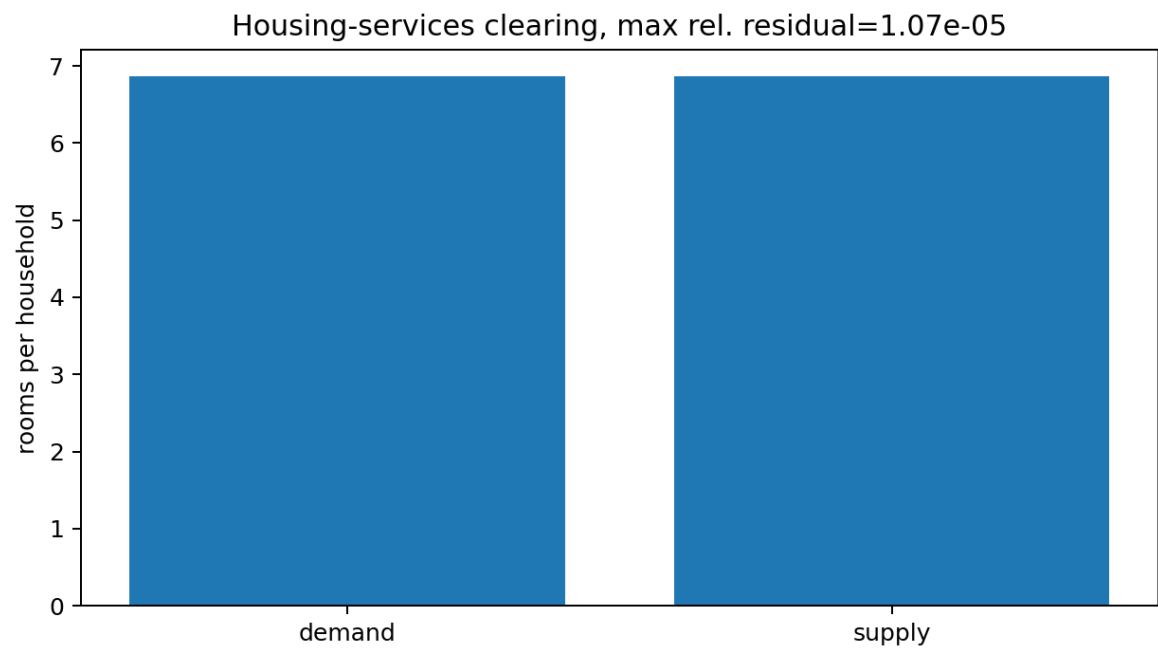
Appendix: D_floor original standard diagnostics (3-4 of 17)

housing by age income state



Original diagnostic image retained unchanged.

housing market



Original diagnostic image retained unchanged.

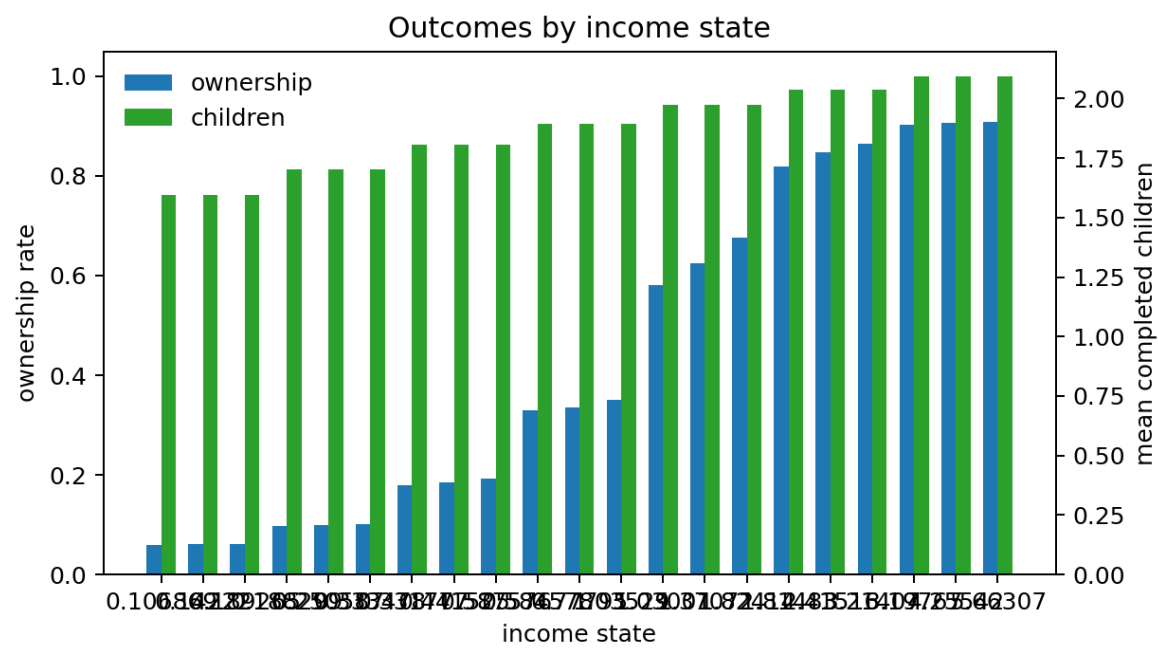
Appendix: D_floor original standard diagnostics (5-6 of 17)

housing prices



Original diagnostic image retained unchanged.

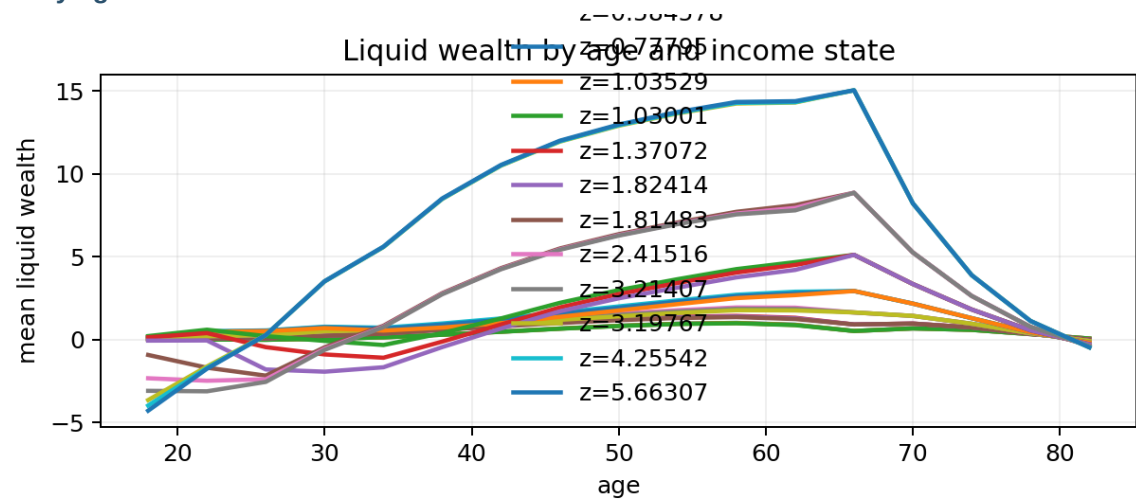
income state outcomes



Original diagnostic image retained unchanged.

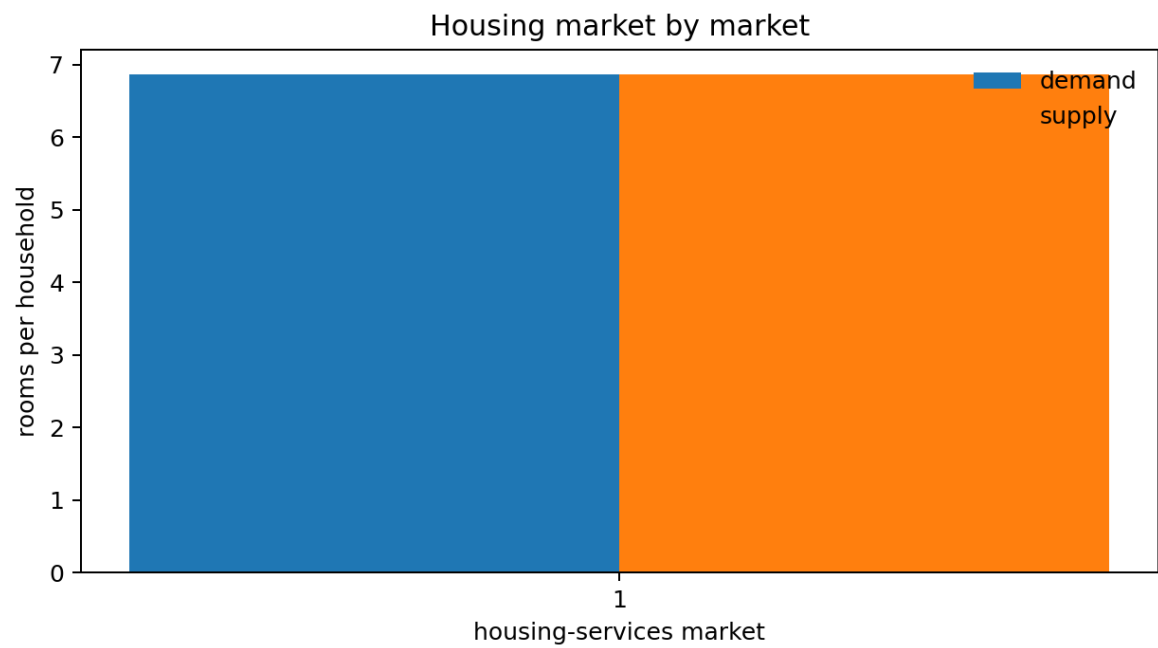
Appendix: D_floor original standard diagnostics (7-8 of 17)

liquid wealth by age income state



Original diagnostic image retained unchanged.

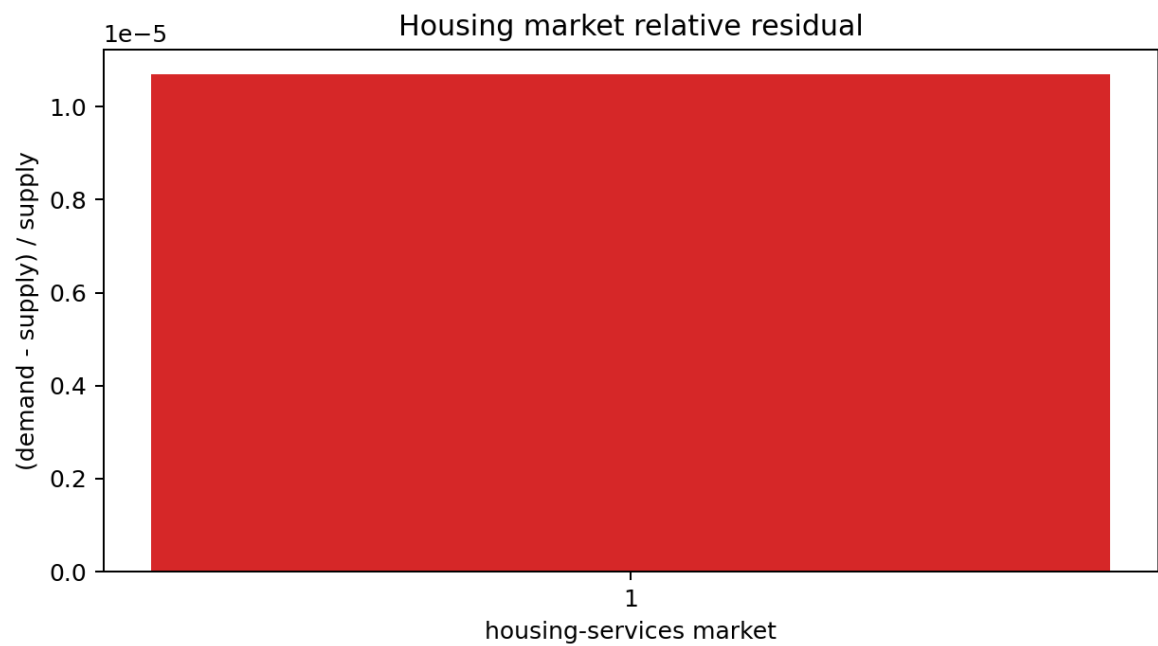
market clearing by market



Original diagnostic image retained unchanged.

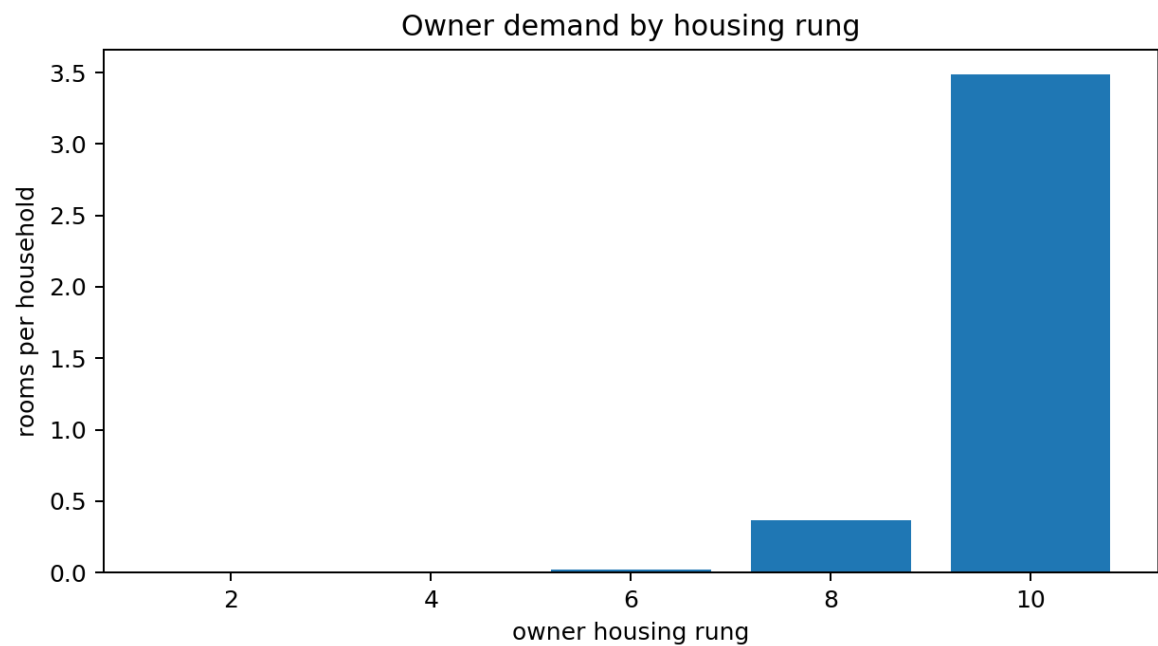
Appendix: D_floor original standard diagnostics (9-10 of 17)

market clearing residuals



Original diagnostic image retained unchanged.

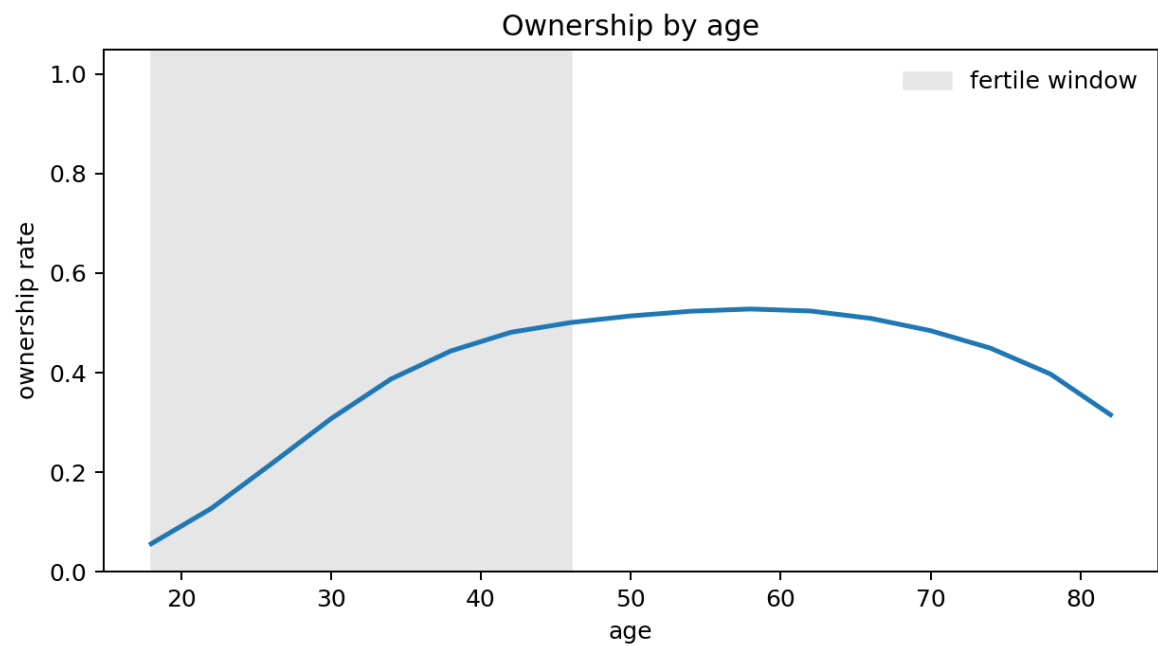
owner rungs



Original diagnostic image retained unchanged.

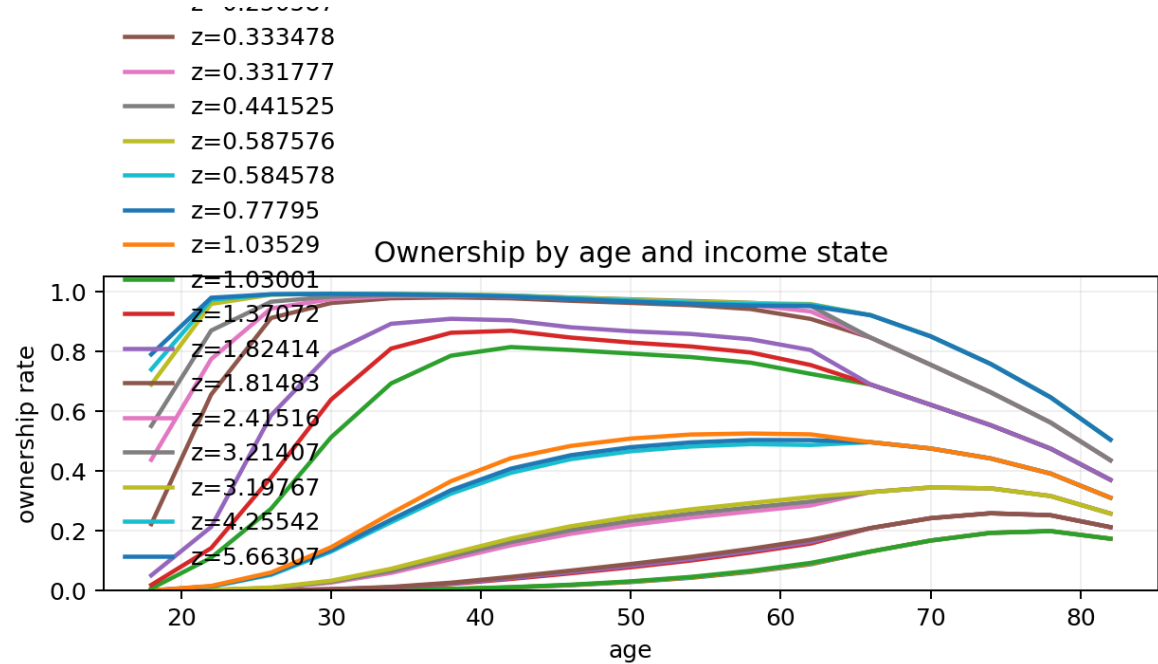
Appendix: D_floor original standard diagnostics (11-12 of 17)

ownership by age



Original diagnostic image retained unchanged.

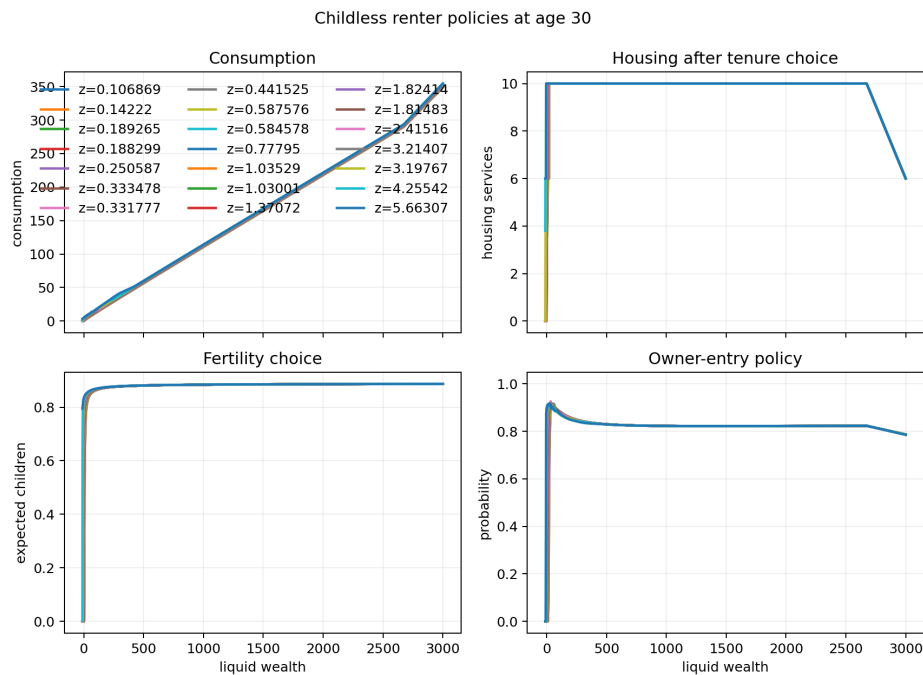
ownership by age income state



Original diagnostic image retained unchanged.

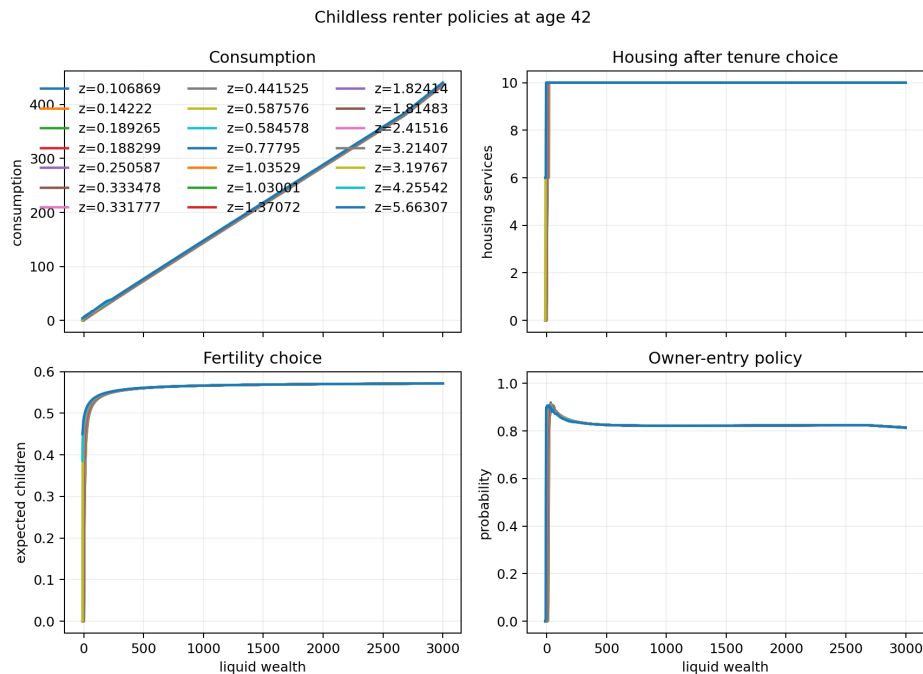
Appendix: D_floor original standard diagnostics (13-14 of 17)

policy childless renter age30



Original diagnostic image retained unchanged.

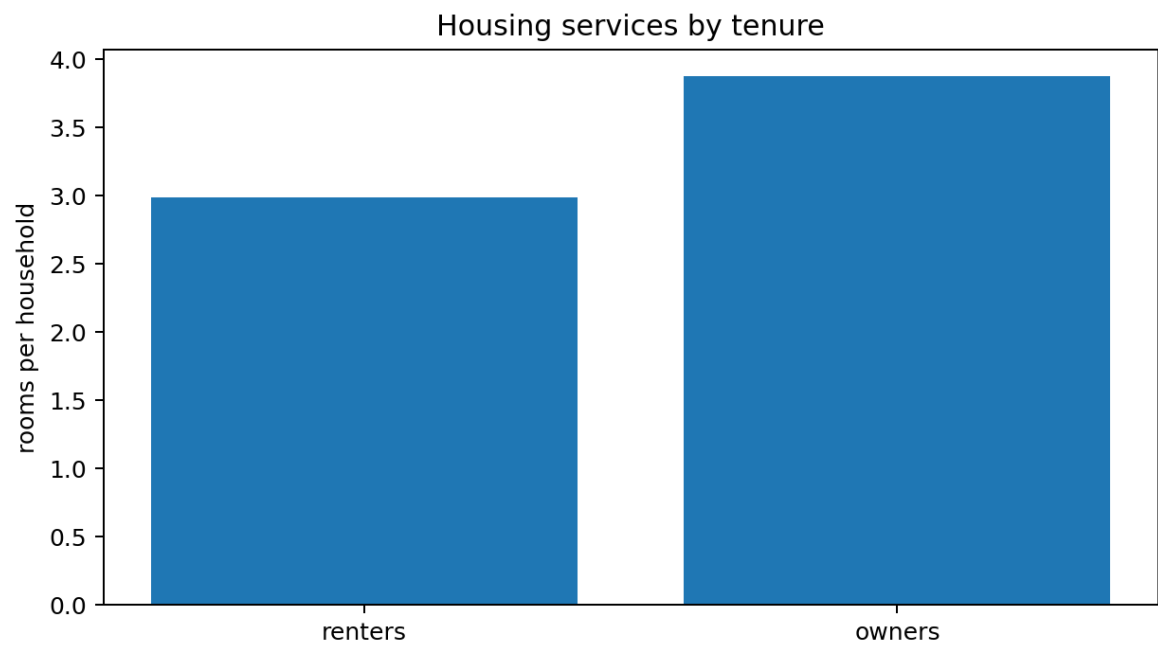
policy childless renter age42



Original diagnostic image retained unchanged.

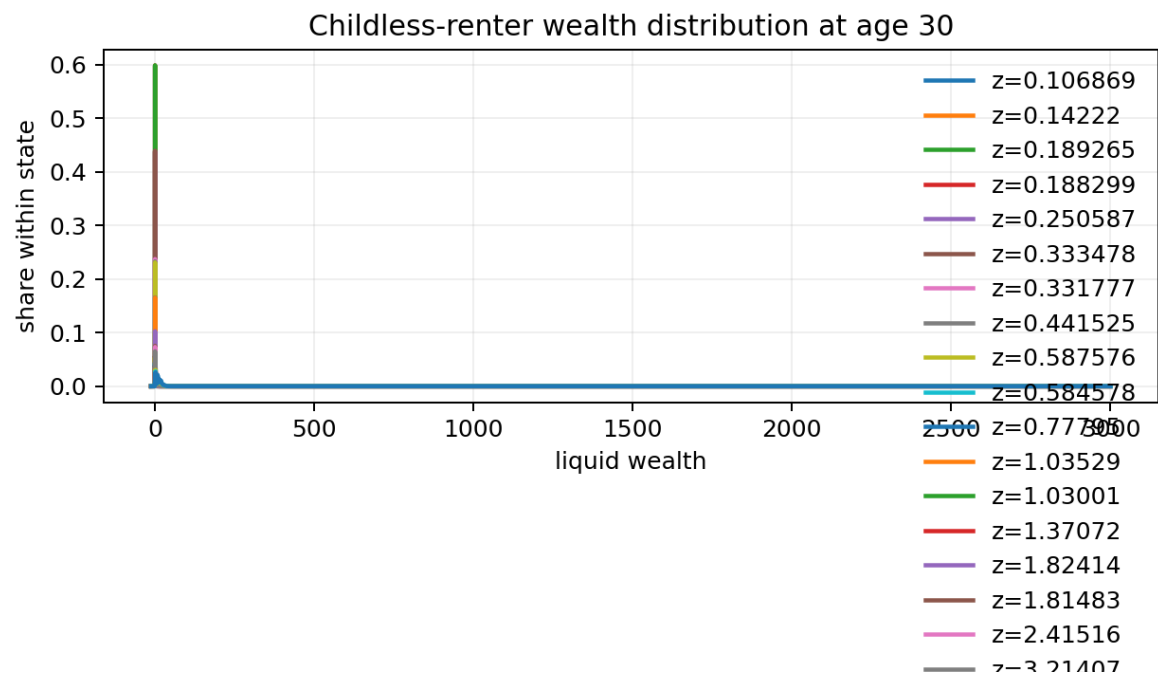
Appendix: D_floor original standard diagnostics (15-16 of 17)

tenure services



Original diagnostic image retained unchanged.

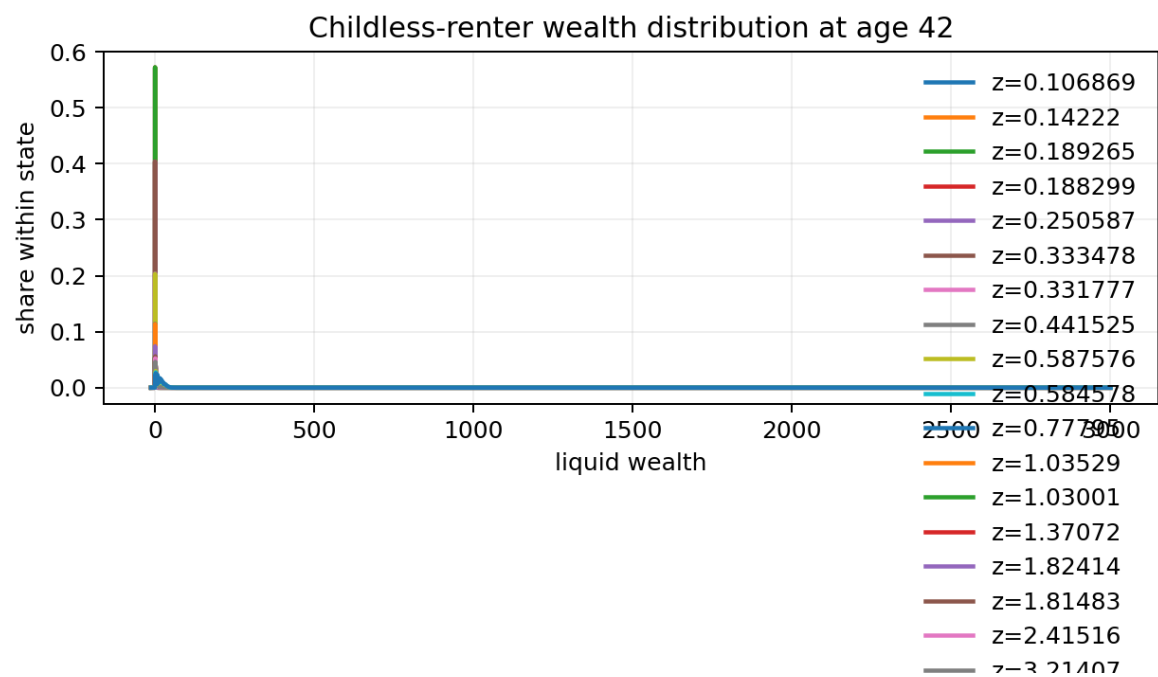
wealth dist childless renter age30



Original diagnostic image retained unchanged.

Appendix: D_floor original standard diagnostics (17-17 of 17)

wealth dist childless renter age42



Original diagnostic image retained unchanged.

Appendix: D_shares full selected target fit

Collector-selected stage: production; case: worker39_proposal18; loss: 1,524.04. Collection verification receipt checkpoint SHA-256: a195eef80e565dbf3ca0b65bd56ffc1ecdc9ba24a39118eb1b3855c2286bc2e9.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Initial fertility normalization	2.1	2.1002	0.00019	-	-
Childless women, ages 40-44	0.19828	0.12597	-0.072309	35,532.30	185.78
Exactly one child among mothers, ages 40-44	0.21366	0.28478	0.071122	26,952.82	136.34
Period mean first-birth age	25.976	26.246	0.26962	139.83	10.165
First births at age 30+	0.24928	0.24595	-0.0033309	13,866.07	0.15384
Wealth / annual gross labor earnings	6.1459	5.1972	-0.94868	7.5951	6.8356
Annual bequests / aggregate wealth	0.0088	0.0063029	-0.0024971	5,165,289.26	32.209
Old wealth/income p90 / median, ages 76-84	3.5159	4.3755	0.85955	10.616	7.8437
Mean occupied rooms, capped at 9	5.5611	6.1667	0.60564	128.02	46.958
Ownership, heads 30-55	0.64833	0.36679	-0.28154	2,339.36	185.43
First-birth room response, -1 to +3	0.72025	0.37748	-0.34276	137.57	16.162
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.54813	0.20106	280.53	11.34
Recent-parent ownership gap	0.1629	-0.017945	-0.18084	27,055.82	884.82

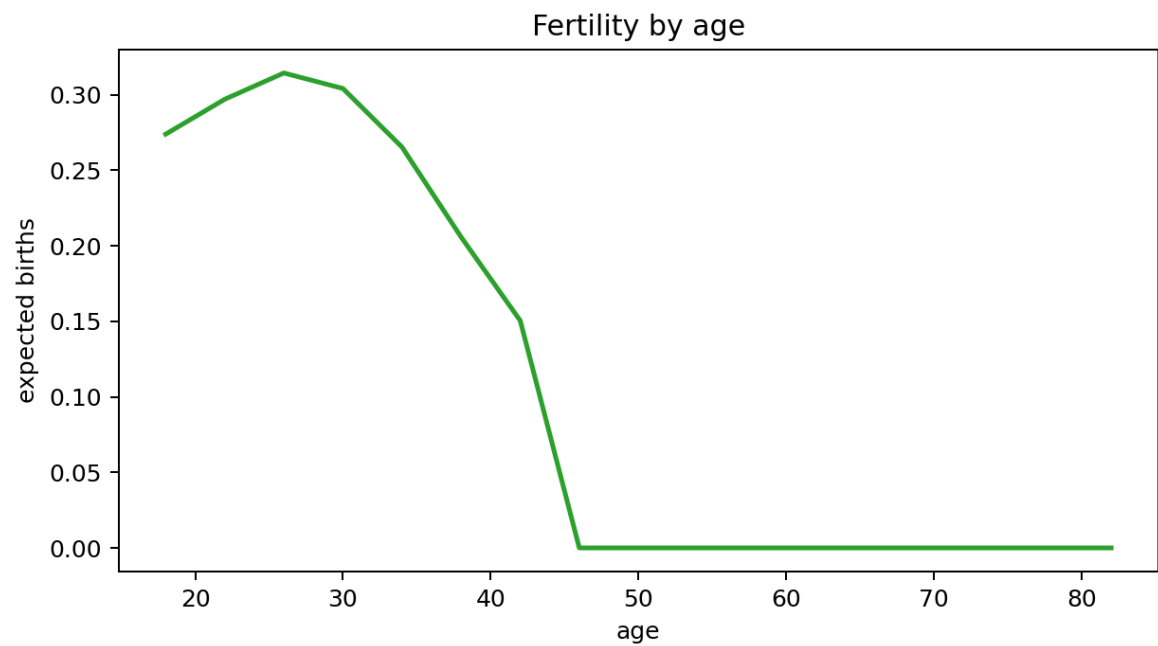
All 19 parameter estimates, bounds, and status

Bounds are taken from the collector's parameters_actual_bounds.csv. 'externally_fixed_or_derived' indicates the parameter is not a searched coordinate in this plan; a missing bound is shown as unavailable.

Parameter	Estimate	Actual lower	Actual upper	Near bound	Status
beta_annual	0.97871	0.94	0.99	False	searched coordinate
kappa_fert	0.28304	0.02	50	True	searched coordinate
kappa_fert_continuation	0.39254	0.02	50	True	searched coordinate
chi	0.98861	0.1	5	False	searched coordinate
H0	7.9218	0.2	80	False	searched coordinate
theta0	0.036824	0	8	True	searched coordinate
theta1	0.091012	0.02	16	True	searched coordinate
first_birth_fixed_cost	0.35074	0	8	False	searched coordinate
delta_alpha_jump	0.0060519	0	0.25	False	searched coordinate
delta_alpha	0.014132	0	0.25	False	searched coordinate
hbar_first_child_jump	0	-	-	-	externally_fixed_or_derived
hbar_child_rooms	0	-	-	-	externally_fixed_or_derived
psi_child	0.061247	-	-	-	externally_fixed_or_derived
payroll_tax	0.179	-	-	-	externally_fixed_or_derived
pension_period	2.0464	-	-	-	externally_fixed_or_derived
housing_supply_elasticity	0.63	-	-	-	externally_fixed_or_derived
tenure_choice_kappa	0.005	-	-	-	externally_fixed_or_derived
alpha_cons	0.733	-	-	-	externally_fixed_or_derived
sigma	2	-	-	-	externally_fixed_or_derived

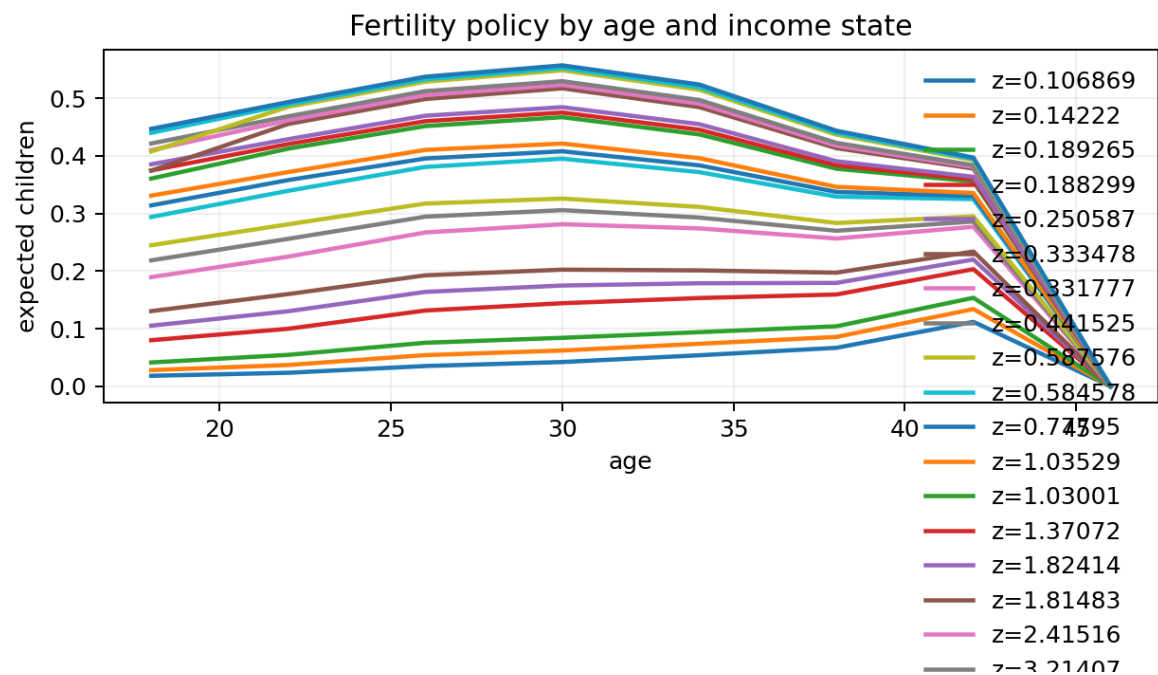
Appendix: D_shares original standard diagnostics (1-2 of 17)

fertility by age



Original diagnostic image retained unchanged.

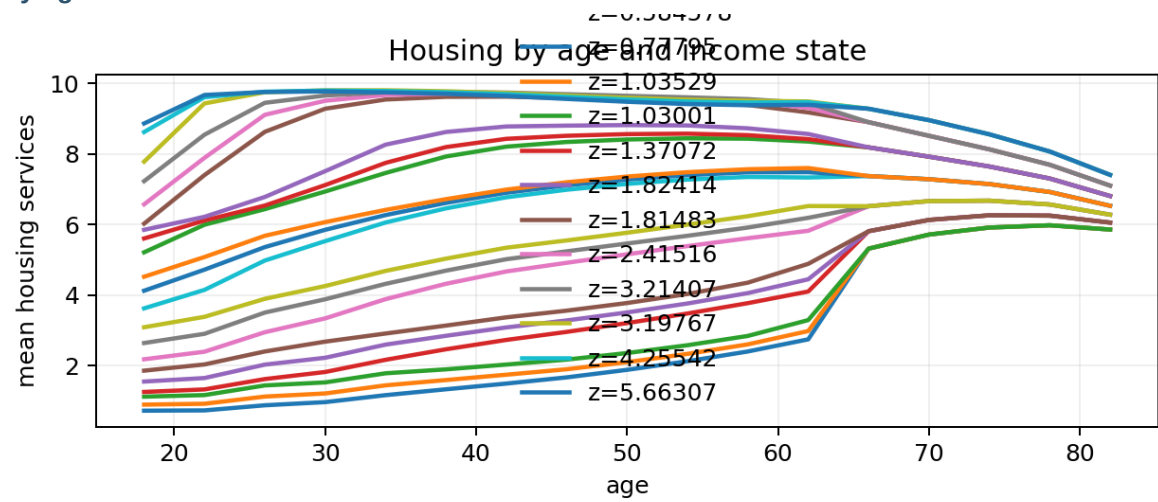
fertility policy by age income state



Original diagnostic image retained unchanged.

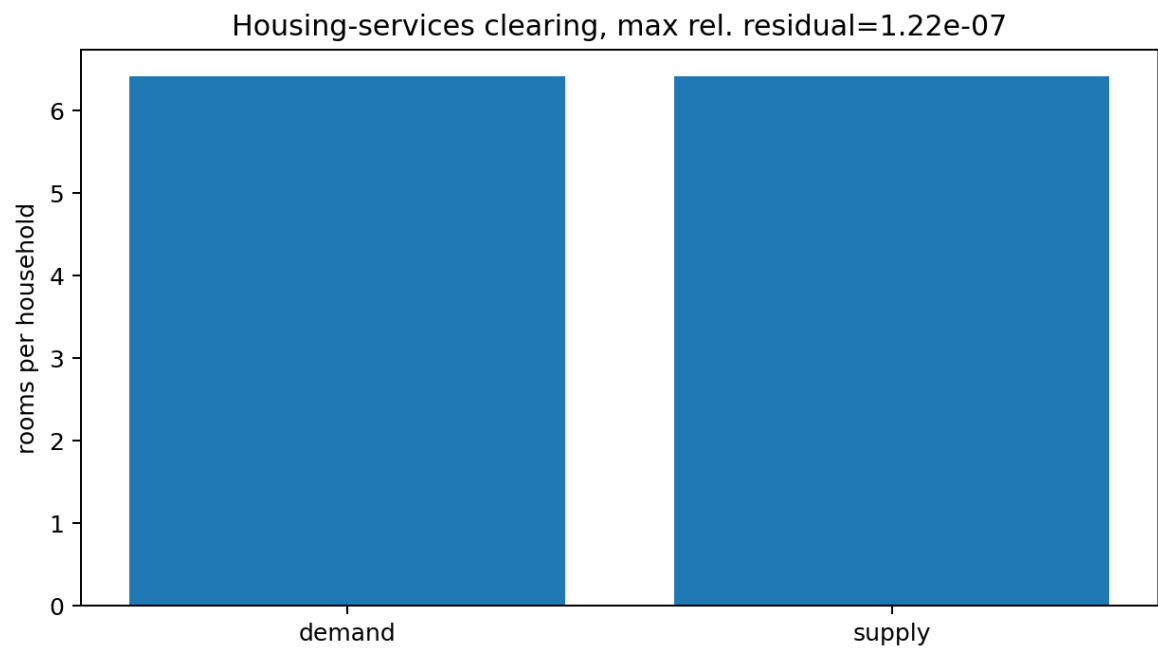
Appendix: D_shares original standard diagnostics (3-4 of 17)

housing by age income state



Original diagnostic image retained unchanged.

housing market



Original diagnostic image retained unchanged.

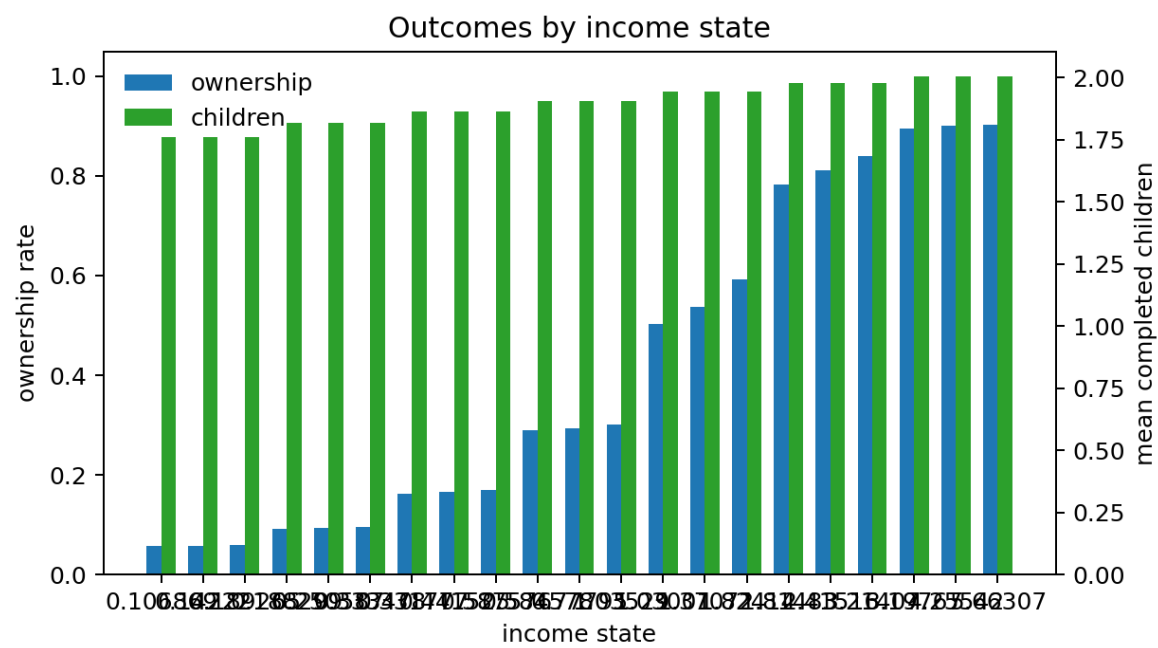
Appendix: D_shares original standard diagnostics (5-6 of 17)

housing prices



Original diagnostic image retained unchanged.

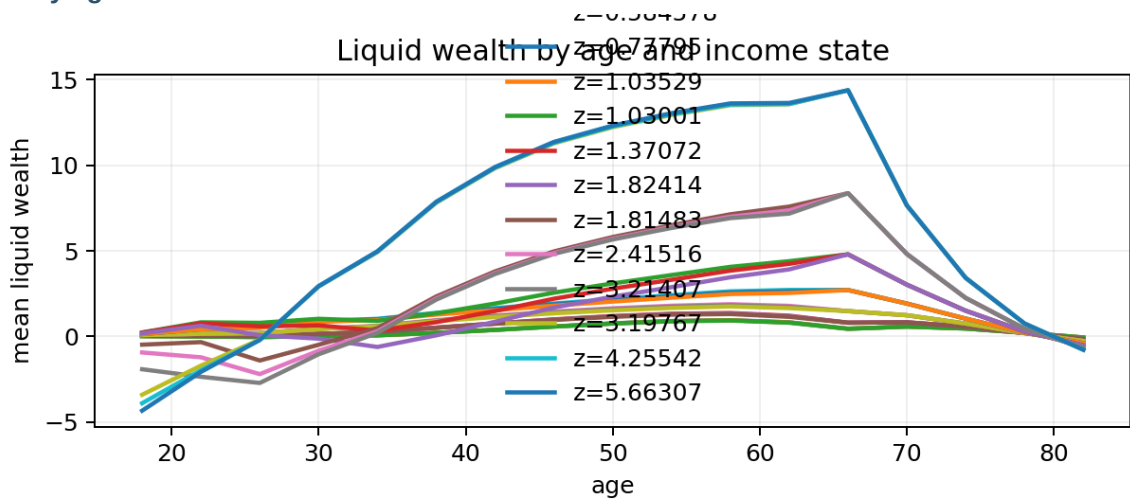
income state outcomes



Original diagnostic image retained unchanged.

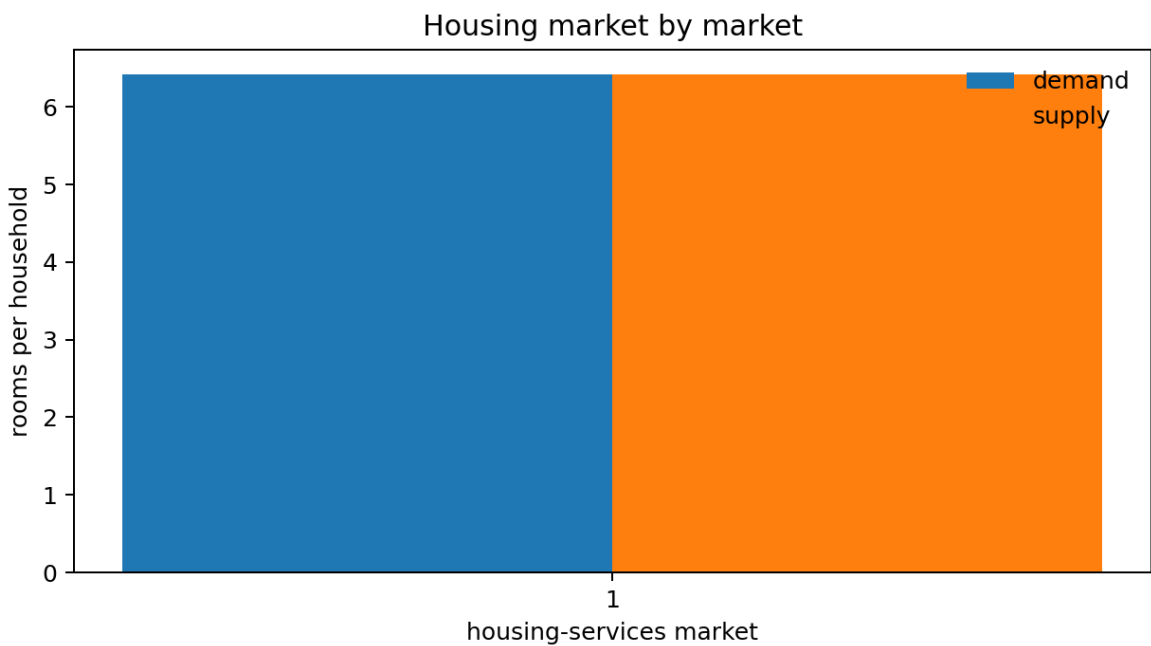
Appendix: D_shares original standard diagnostics (7-8 of 17)

liquid wealth by age income state



Original diagnostic image retained unchanged.

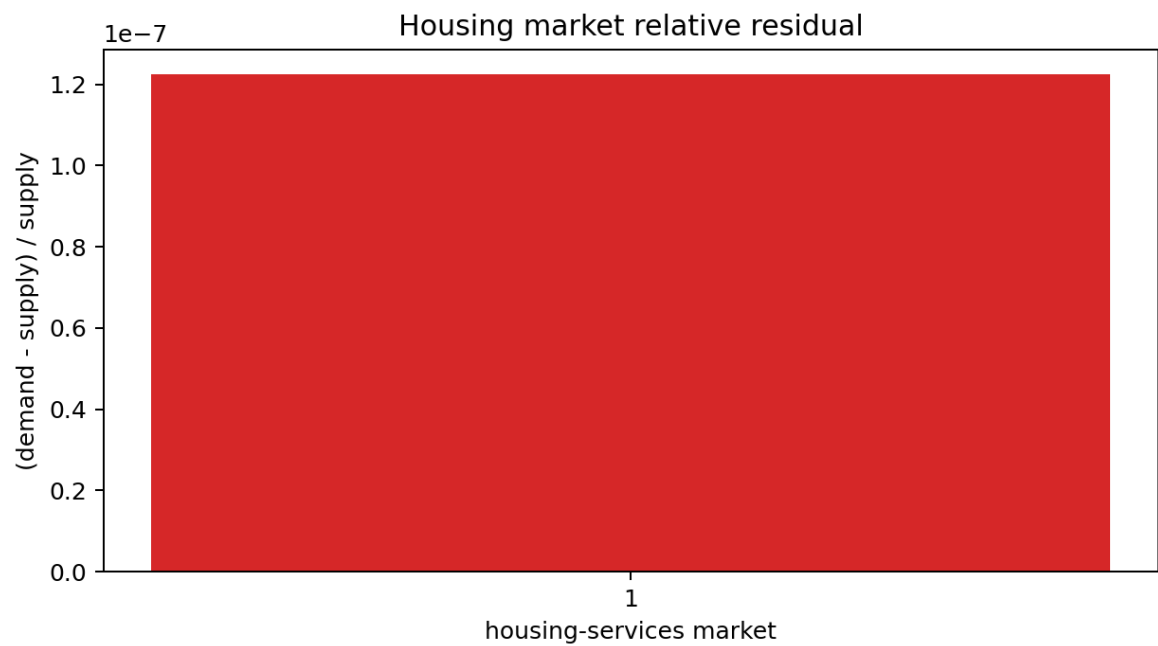
market clearing by market



Original diagnostic image retained unchanged.

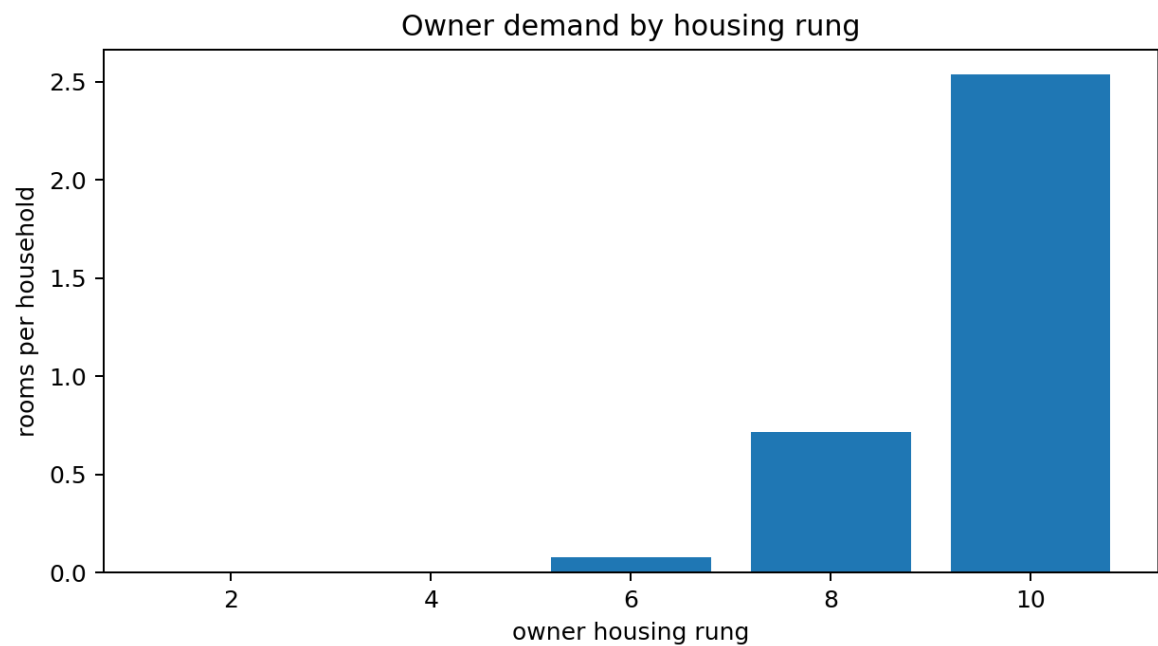
Appendix: D_shares original standard diagnostics (9-10 of 17)

market clearing residuals



Original diagnostic image retained unchanged.

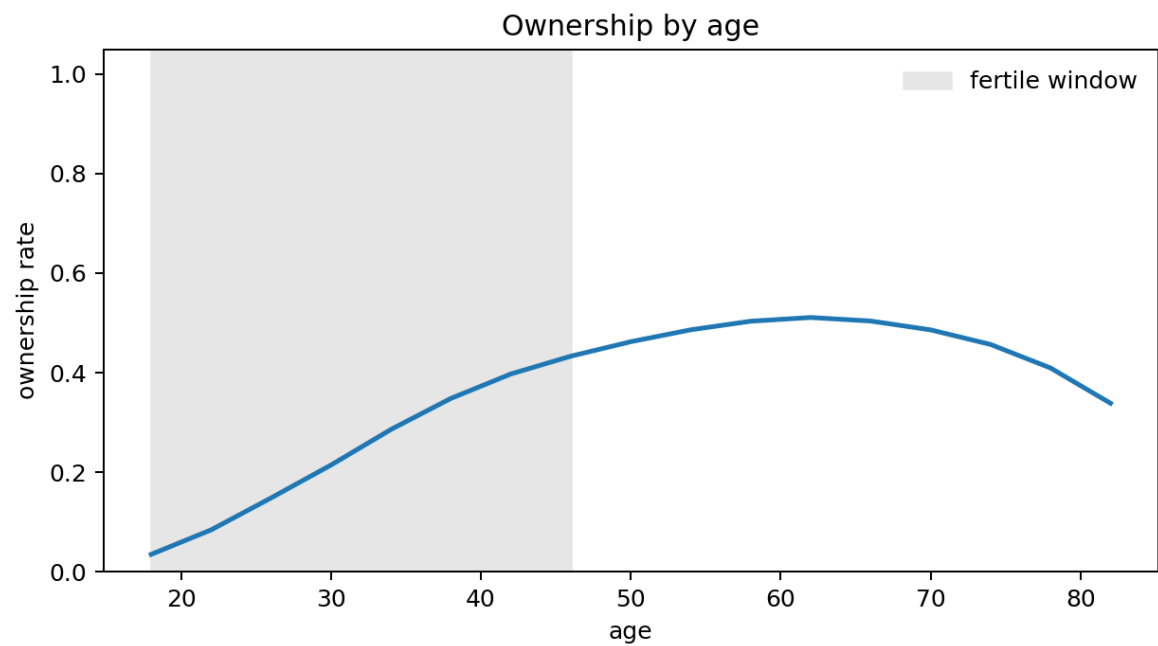
owner rungs



Original diagnostic image retained unchanged.

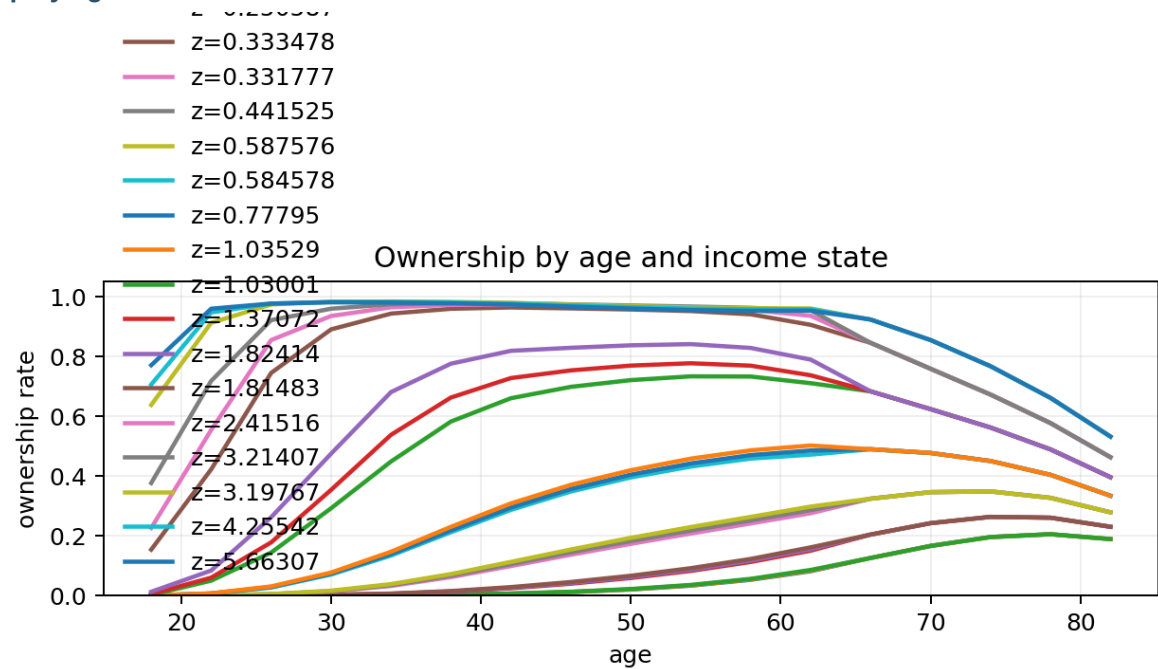
Appendix: D_shares original standard diagnostics (11-12 of 17)

ownership by age



Original diagnostic image retained unchanged.

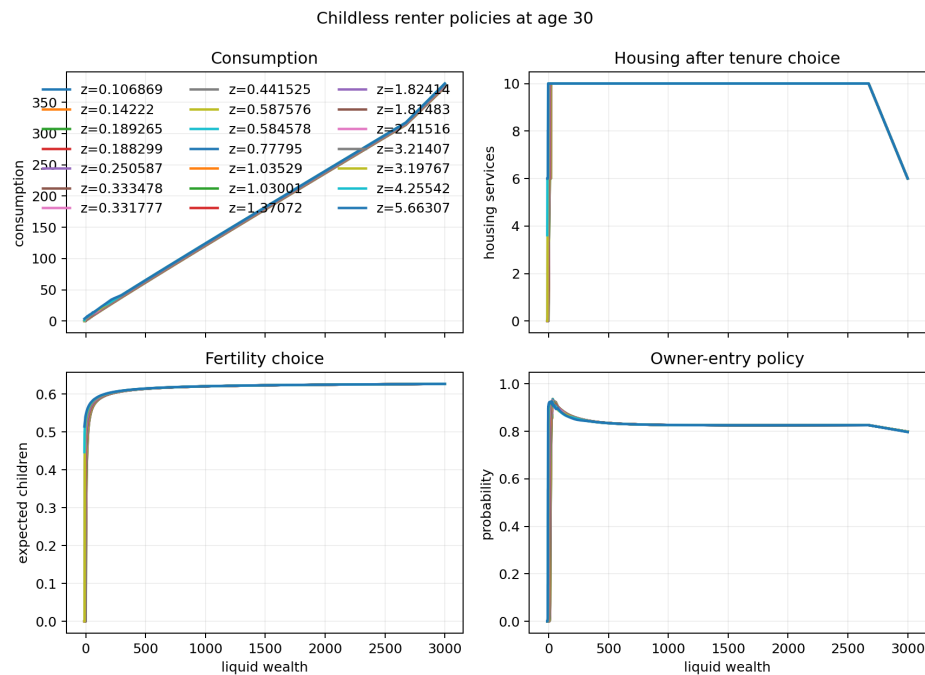
ownership by age income state



Original diagnostic image retained unchanged.

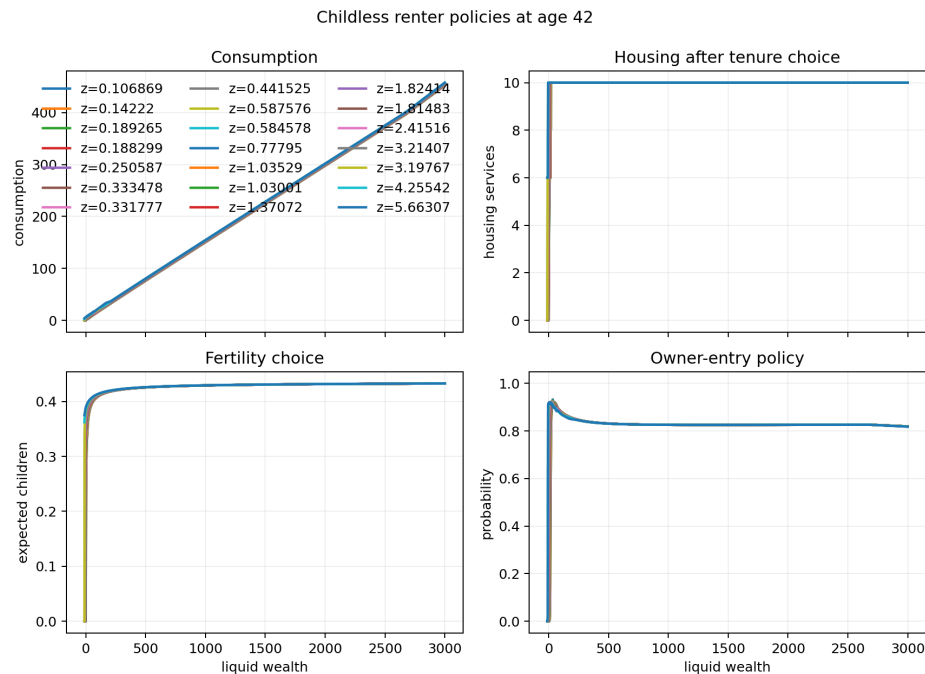
Appendix: D_shares original standard diagnostics (13-14 of 17)

policy childless renter age30



Original diagnostic image retained unchanged.

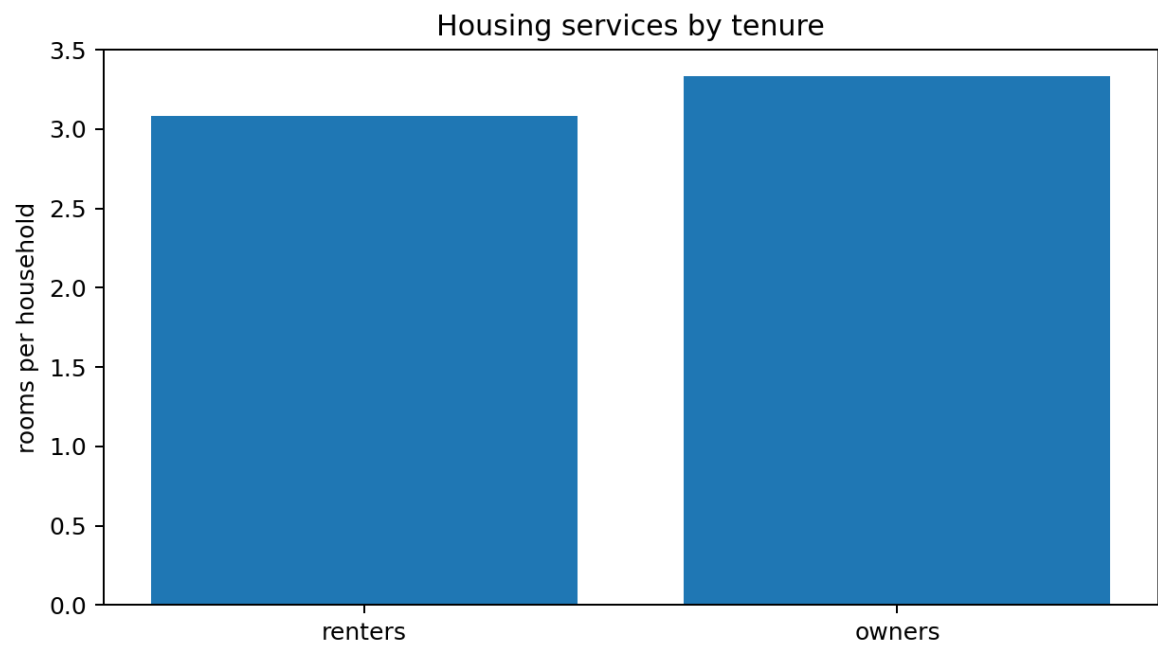
policy childless renter age42



Original diagnostic image retained unchanged.

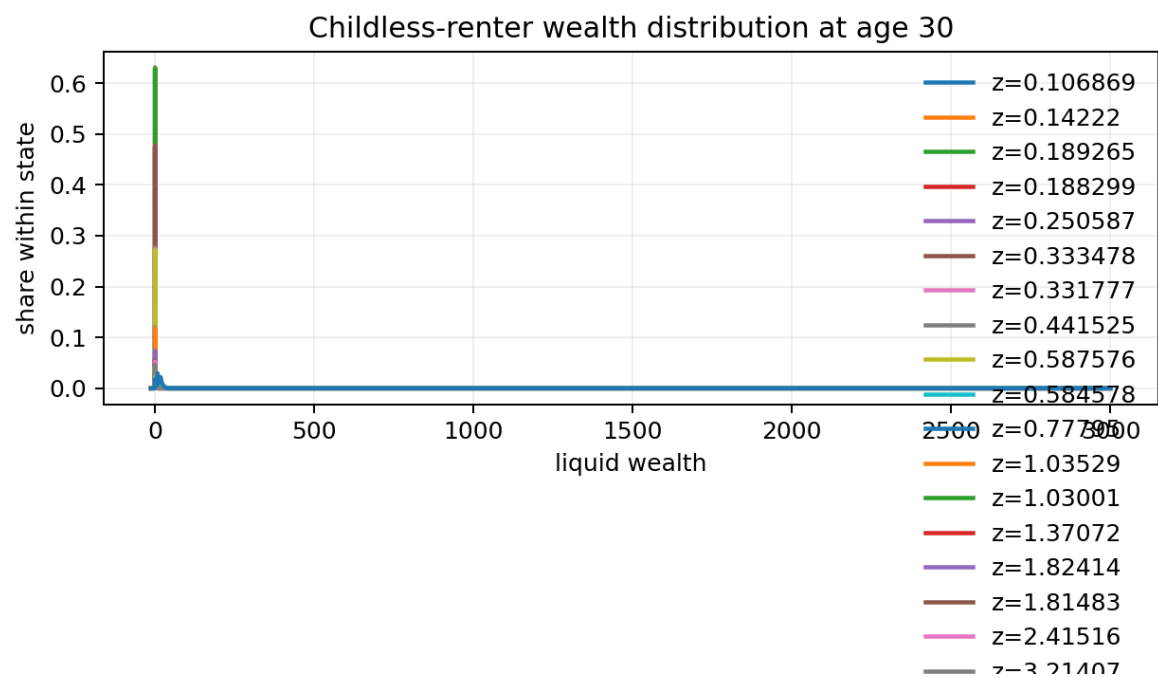
Appendix: D_shares original standard diagnostics (15-16 of 17)

tenure services



Original diagnostic image retained unchanged.

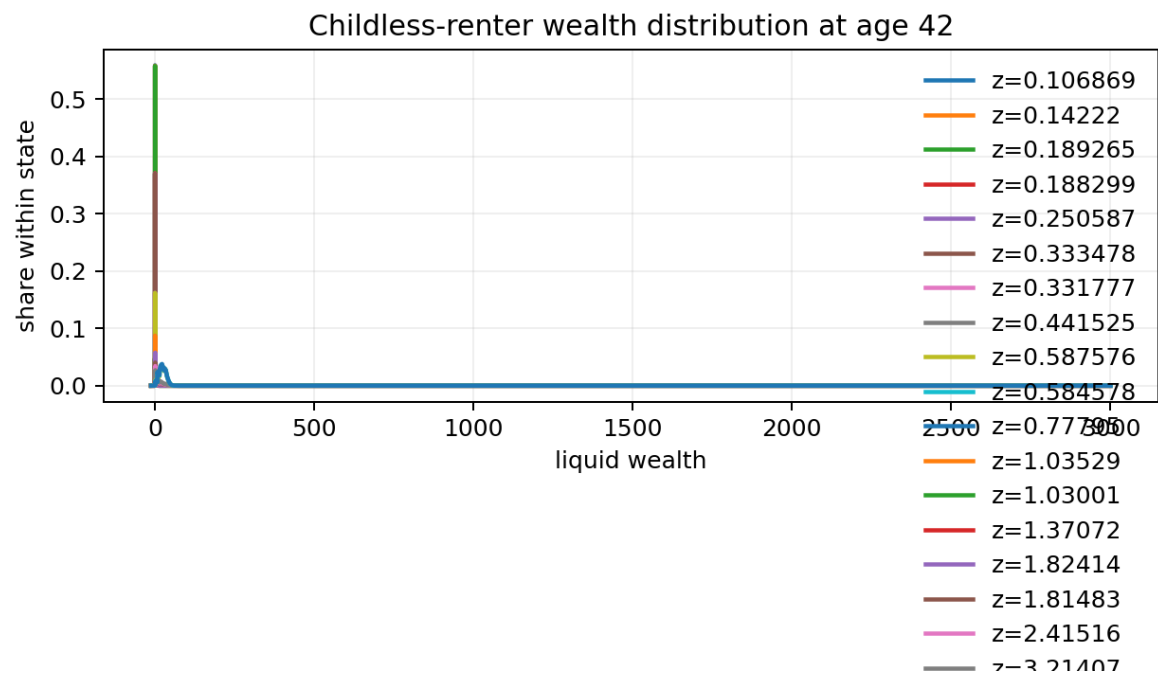
wealth dist childless renter age30



Original diagnostic image retained unchanged.

Appendix: D_shares original standard diagnostics (17-17 of 17)

wealth dist childless renter age42



Original diagnostic image retained unchanged.

Sources

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